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supervisé par
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Green-Function Approaches to Lattice Vibrations in Disordered Materials

Rapporteurs:

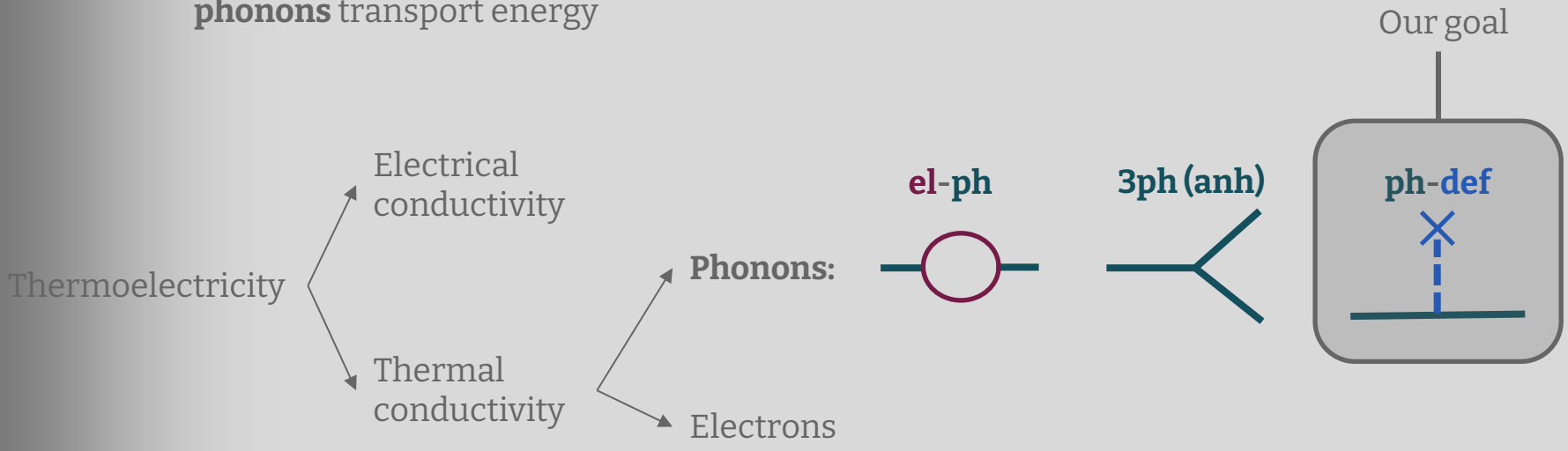
- **Matteo Cococcioni** — University of Pavia
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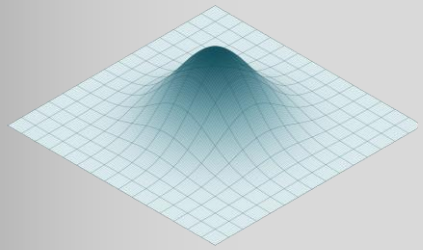
- **Konstantinos Termentzidis** — INSA, Lyon
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- **François-Xavier Coudert** — IRCP, Université PSL

Outline

- Role of **defects**:
  - thermoelectricity
  - semiconductor physics
  - alloying
- *Ab Initio* **electrical** and **thermal conductivity**
- Thermal conductivity depends on how efficiently **electrons** and **phonons** transport energy



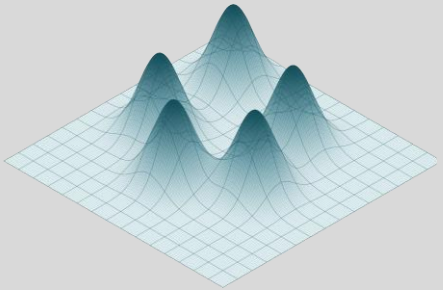
Outline



Ab Initio
single-defect
local potential

Consistent
single-defect
approximation

Multi-defect
solution



My contributions

Numerically **stable**
spectral functions



Thermal Transport of lattice vibrations

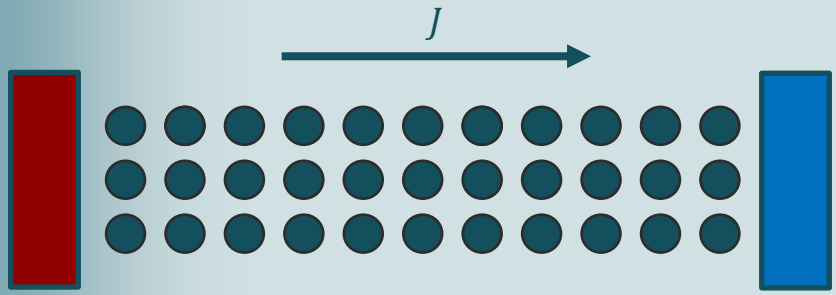
1.1 Thermal transport
1.2 Phonons
1.3 Perturbation theory

Thermal transport

- How does heat propagate through a solid?

$$J = -\kappa \nabla T$$

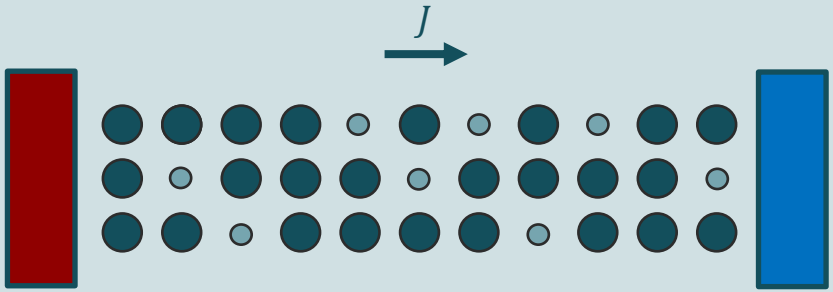
J: thermal flux
κ: **thermal conductivity**
∇T: thermal gradient



Lattice perspective

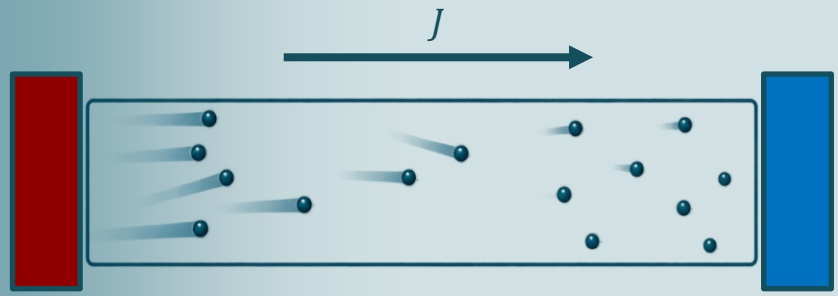
- How is this transport modified by **disorder**?

It **degrades** thermal conductivity κ



Thermal transport

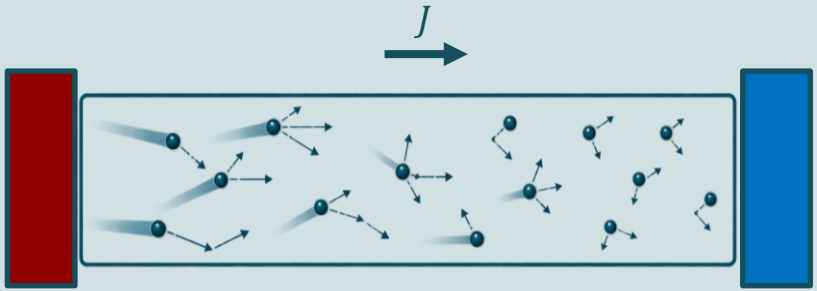
- κ → how efficiently energy **carrier** propagate



Carrier perspective

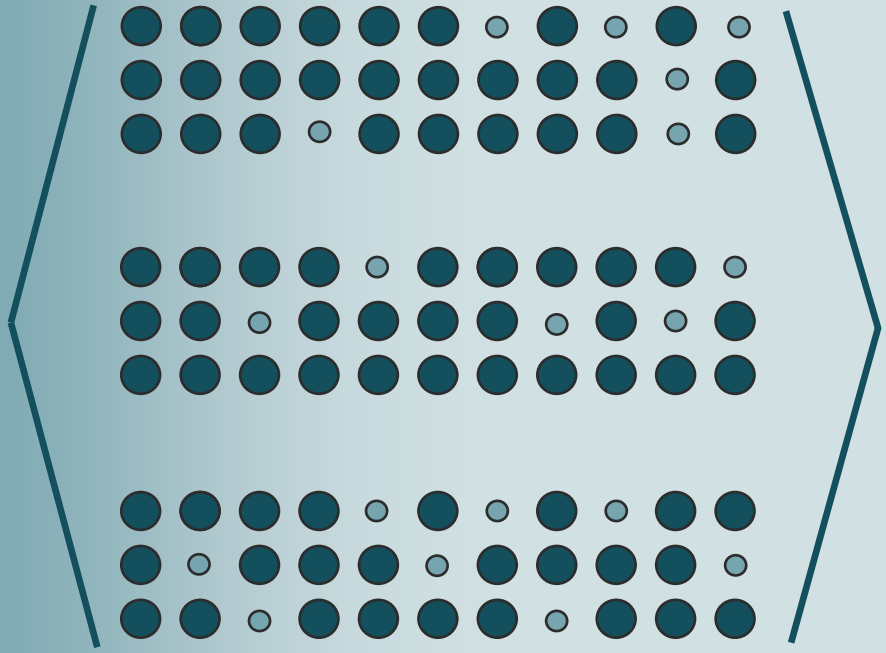
- $\kappa \sim l \sim v\tau$

l: scattering length
v: velocity
τ: **lifetime**



Thermal transport

- Many **configurations** → $\langle \kappa \rangle$



- Calculating the average is **difficult**:
  - high number of configurations
  - expensive *ab initio* calculation

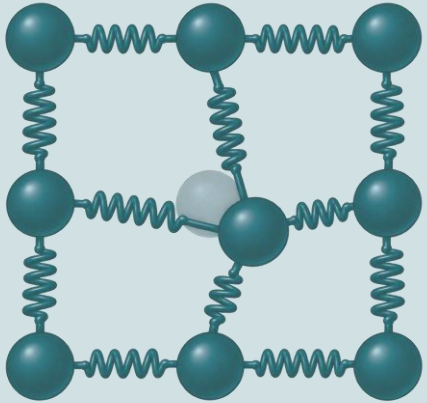
Phonons

- Lattice vibrations (**phonons**) are energy carriers
- Ionic displacements around **equilibrium** position → **harmonic** approximation

$$H = \sum_i \frac{p_i^2}{2M_i} + \frac{1}{2}\sum_{ij} \Phi_{ij} u_i u_j + \mathcal{O}(u^3)$$

u: ionic displacement
p: ionic momentum
M: ionic mass

- **Interatomic Force Constants** (IFC) $\Phi = \partial^2 E / \partial u^2$



Phonons

- **Phonons** are eigenvectors of Newton's equation:

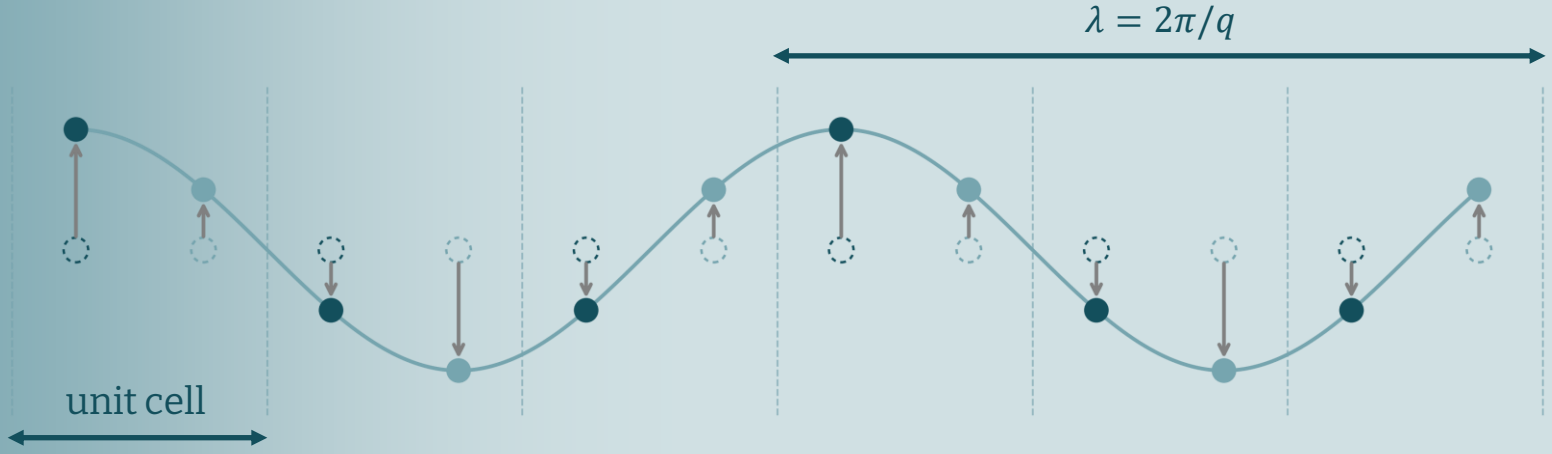
$$\mathrm{M}\omega^2 u = \Phi u$$

$$\omega^2 u = \frac{\Phi}{M} u := Du$$

D: dynamical matrix

Stationary states → **infinite** lifetime

- **Translational Invariance** $\Phi(\boldsymbol{R}, \boldsymbol{R}') = \Phi(\boldsymbol{R} - \boldsymbol{R}', \boldsymbol{0})$ makes Φ diagonal in q-space



Perturbation theory

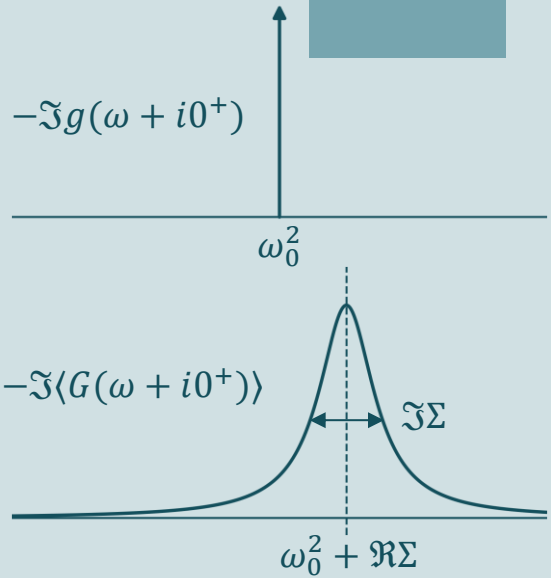
- **Disorder** configuration $\alpha \rightarrow$ new M^α and Φ^α

$$(M_0 + \Delta M^\alpha)\omega^2 u = (\Phi_0 + \Delta\Phi^\alpha)u$$
$$\omega^2 u = (D_0 + V^\alpha)u$$

Scattering **potential** $M_0 V^\alpha(\omega) = \Delta\Phi^\alpha - \Delta M^\alpha \omega^2$

- **Green's functions**:
  - unperturbed $\rightarrow$ $g(z) = (z^2 - D_0)^{-1}$
  - single configuration $\rightarrow$ $G^\alpha(z) = (z^2 - D_0 - V^\alpha)^{-1}$
  - average $\rightarrow$ $\langle G(z)\rangle = (z^2 - D_0 - \Sigma(z))^{-1}$

Self-energy $\Sigma(z)$ is the **effective** correction after average



Perturbation theory

$$u(t) \sim e^{i\omega t} \sim e^{i\Omega t} e^{-\omega_0^{-1}\Im\Sigma t}$$

- Σ is **complex**:

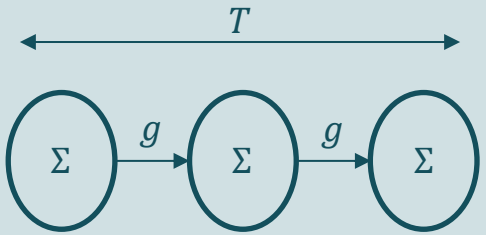
$$\Omega^2 = \omega_0^2 + \Re\Sigma(\omega)$$
$$\tau^{-1} = \omega_0^{-1}\Im\Sigma(\omega)$$

Ω: renormalized frequency
τ: lifetime

- If V^α represents the **single** scattering event

$$T^\alpha = V^\alpha + V^\alpha g V^\alpha + \cdots = V^\alpha(1 - gV^\alpha)^{-1}$$
$$\langle T^\alpha \rangle = \Sigma + \Sigma g \Sigma + \cdots = \Sigma(1 - g\Sigma)^{-1}$$

T-matrix T contains **every** scattering
Σ contains **irreducible** scattering



$$\langle G \rangle = g + g\Sigma\langle G \rangle$$

$$\langle G \rangle = g + g\langle T \rangle g \quad \longleftrightarrow \quad \langle T \rangle = \Sigma + \Sigma g \langle T \rangle$$

Defective Potential

1. Defect force constants
2. Centering
3. Acoustic Sum Rule

Defect force constants

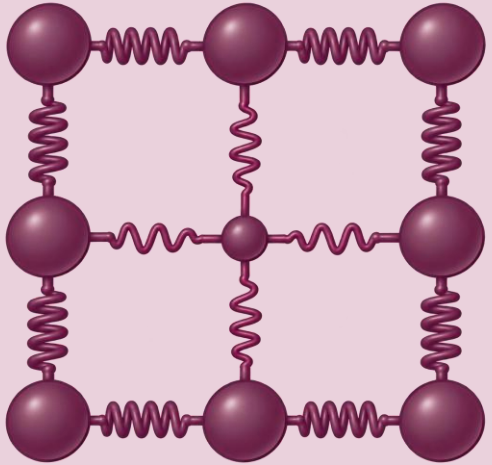
- How we construct V^{α}?
  Starting from V_1

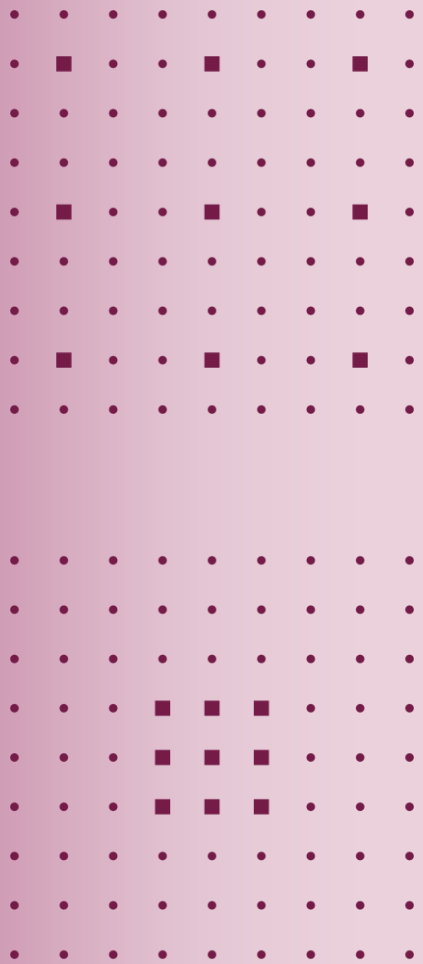
- How a **single** defect perturbs the phonons?

  $\Phi_0 \quad \rightarrow \quad$ **pristine** supercell
  $\Phi \quad \rightarrow \quad$ single **defect**
  from *ab initio* in **small periodic** supercells

- **Perturbing** potential is
  $$M_0 V_1(\omega) = \Delta\Phi_1 - \Delta M_1 \omega^2$$

- How we can make V_1 **local**?







Centering

- Φ_0 is defined over a coarse **periodic** supercell

  But we need phonons in a **fine** q-grid

- Φ_0 depends only on $\boldsymbol{R}_- = \boldsymbol{R} - \boldsymbol{R}'$

$$\Phi_0(\boldsymbol{Q}) \underset{\text{FT}}{\Rightarrow} \Phi_0(\boldsymbol{R}_-) \underset{\text{CNT}}{\Rightarrow} \Phi_0(\boldsymbol{r}) \underset{\text{FT}}{\Rightarrow} \Phi_0(\boldsymbol{q})$$

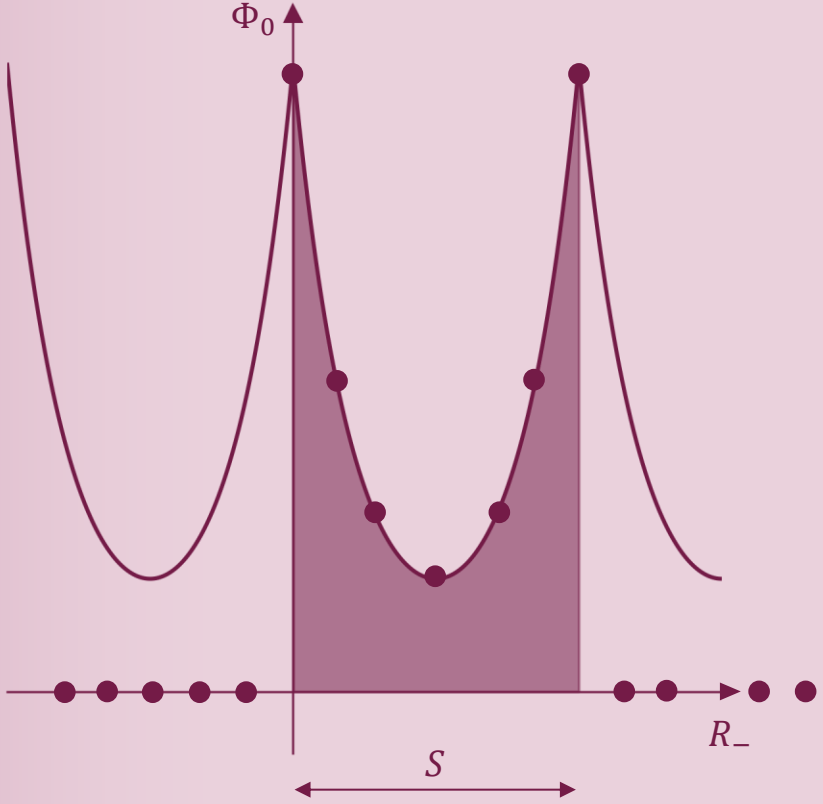
The process is called **Fourier interpolation**

- Φ_0 (and Φ) decay **exponentially** with R_-

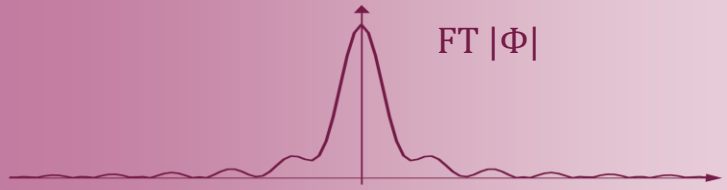




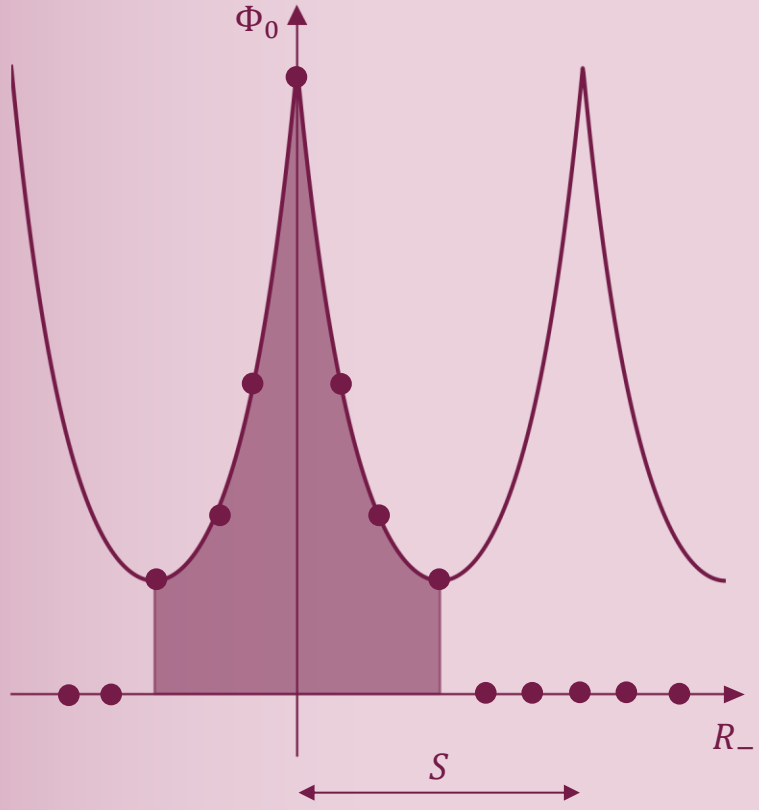
Centering



- **Discontinuity** in R → **oscillations** in q

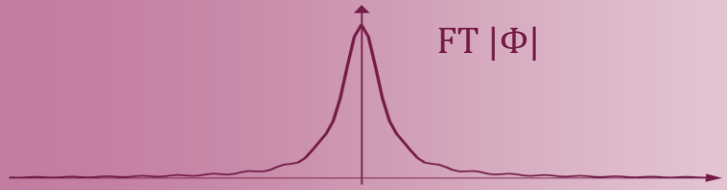


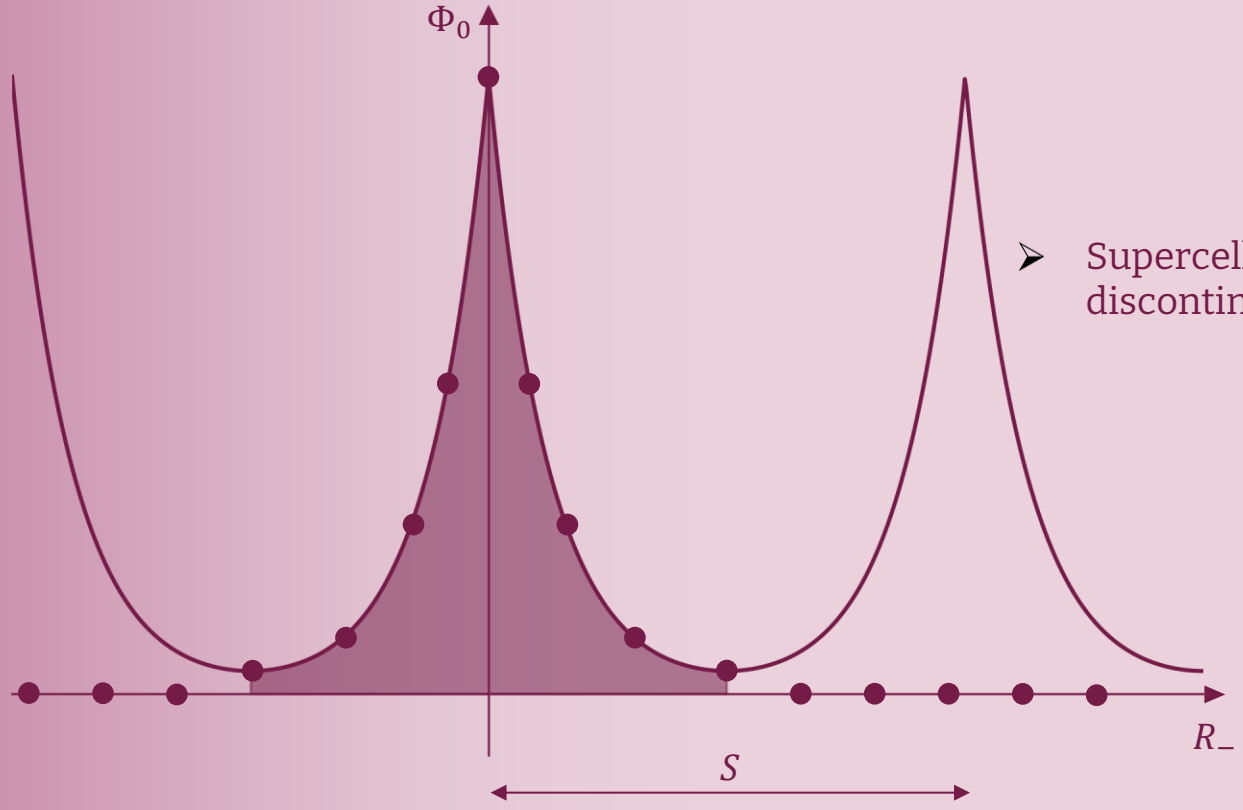
Centering



- Φ is **centered** in $\left[-\frac{S}{2}, \frac{S}{2}\right]$ → discontinuity is **reduced**

Centering





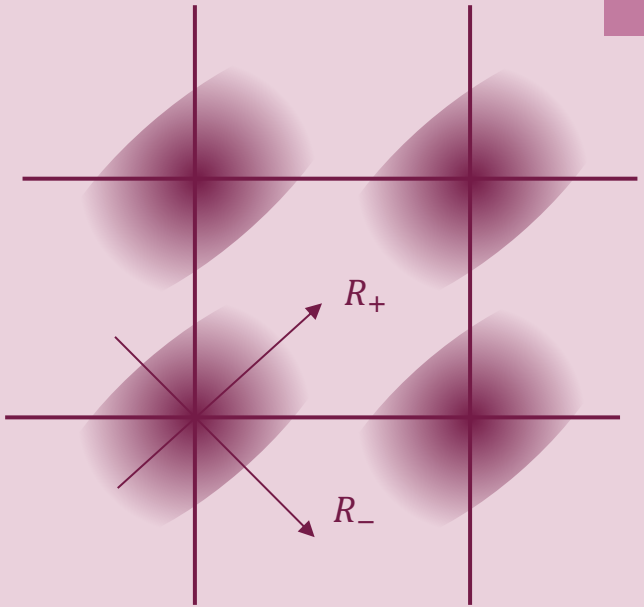
- Supercell **size** increase → discontinuity is **reduced**

Centering

- What about Φ ?
  1. **Fine** q-resolution
  2. **Rigid local** potential

- Defect at $\boldsymbol{P} = \boldsymbol{0}$ breaks translation invariance: $\Phi(\boldsymbol{R}, \boldsymbol{R}')$

- $\Delta\Phi = \Phi - \Phi_0$ decays with $R_+ = ||\boldsymbol{R} + \boldsymbol{R}'||$

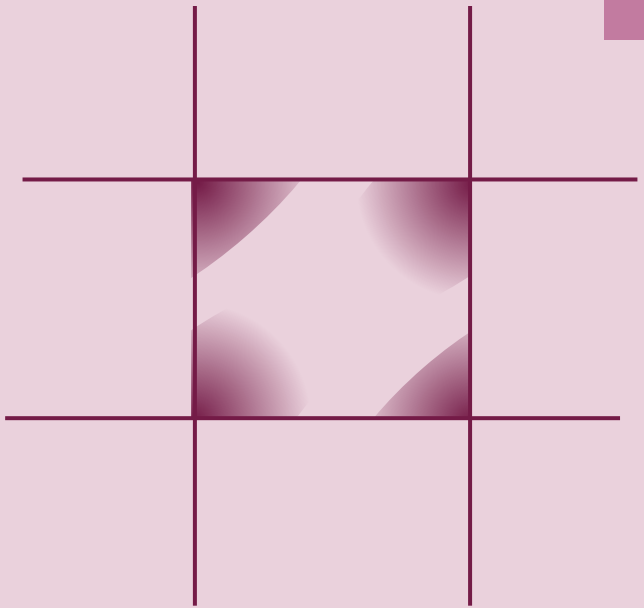


Centering

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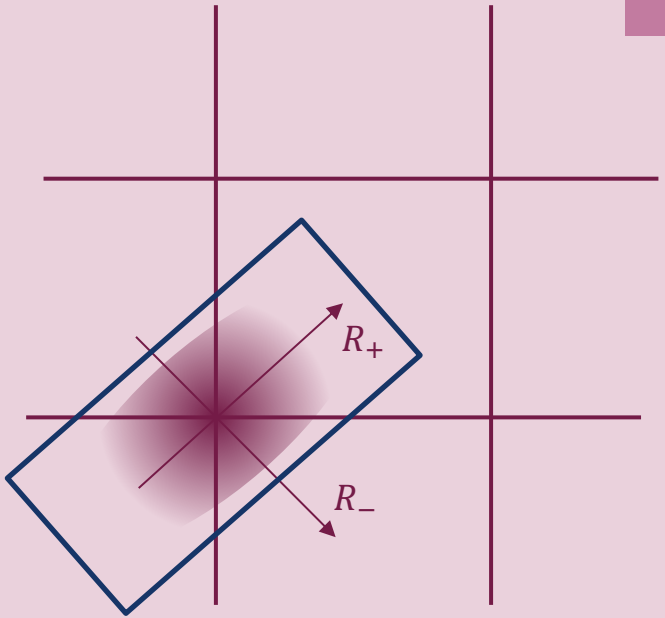
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Centering

- What about Φ ?
  1. **Fine** q-resolution
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- Defect at $\boldsymbol{P} = \boldsymbol{0}$ breaks translation invariance: $\Phi(\boldsymbol{R}, \boldsymbol{R}')$
- $\Delta\Phi = \Phi - \Phi_0$ decays with $R_+ = ||\boldsymbol{R} + \boldsymbol{R}'||$
- Fourier interpolation gives $\Delta\Phi(\boldsymbol{q}, -\boldsymbol{q}')$



Acoustic Sum Rule

- Total energy should be invariant upon **bulk translations**

  $\boldsymbol{q} = \mathbf{0}$, three modes at Γ have **zero** frequency

- Acoustic Sum Rules (ASR) **enforce** this property by changing IFCs:

$$\Phi' B = 0$$

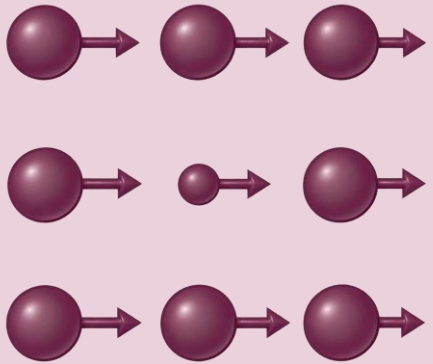
- The rule should preserve **derivative exchange**

$$\Phi'^{T} = \Phi'$$

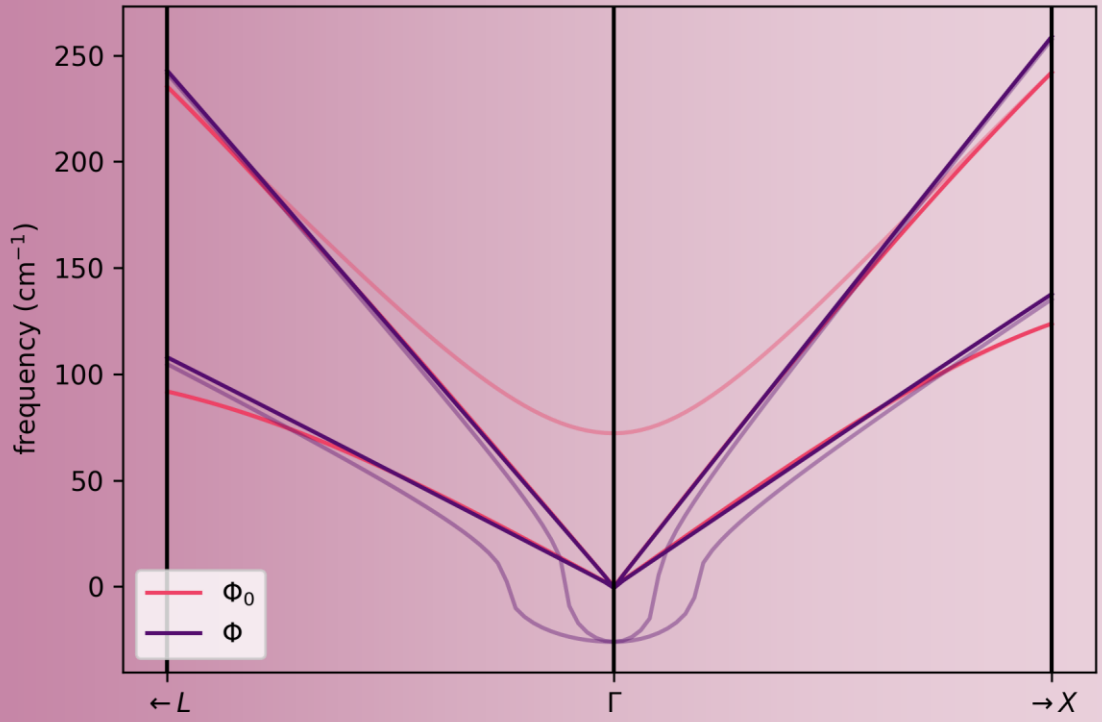
  forces a process and its inverse to have same amplitude

- Our ASR:

$$\min_{\Phi'} ||\Phi' - \Phi||$$



Acoustic Sum Rule



Disorder approximations

3.1 Diagramatic expansion
3.2 Virtual Crystal Approximation (VCA)
3.3 Average T-matrix Approximation (ATA)
3.4 Dynamical Cluster Approximation (DCA)

Diagrammatic expansion

- **Translation** in q-space:

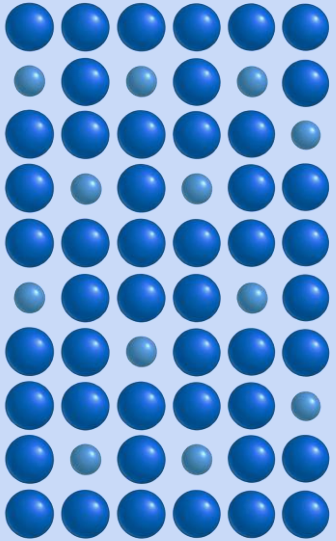
$$V_1(\boldsymbol{q}, -\boldsymbol{q}', \boldsymbol{P}) = e^{-i(\boldsymbol{q}-\boldsymbol{q}')\cdot\boldsymbol{P}} V_1(\boldsymbol{q}, -\boldsymbol{q}', \boldsymbol{0})$$

- **Single** defect:

$$T_1 = V_1 + V_1 g V_1 + \cdots = V_1(1 - gV_1)^{-1}$$

The geometric **inversion** is defined even when the series would diverge.

It's singular at $g(\omega)V_1 = 1$, where **localized states** lie



Diagrammatic expansion

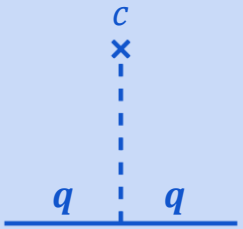
- A real system has a **random** configuration of defects

$$V^{\alpha}(\boldsymbol{q},-\boldsymbol{q}') = \sum_{\boldsymbol{P}\in\alpha} V_1(\boldsymbol{q},-\boldsymbol{q}',\boldsymbol{P}) = V_1(\boldsymbol{q},-\boldsymbol{q}',\boldsymbol{0}) \sum_{\boldsymbol{P}} e^{-i(\boldsymbol{q}-\boldsymbol{q}')\cdot\boldsymbol{P}} \rho_{\alpha}(\boldsymbol{P})$$

- The average T-matrix

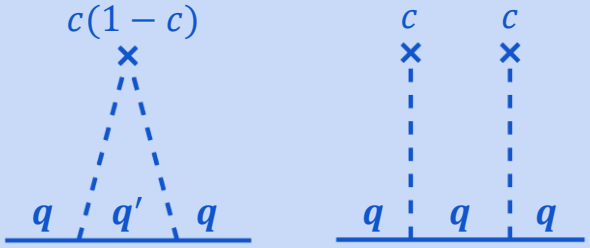
$$\langle T^{\alpha}\rangle = \langle V^{\alpha}\rangle + \langle V^{\alpha} g V^{\alpha}\rangle + \cdots$$

does **not** have a closed solution



First order: $\sum_{\boldsymbol{P}} e^{-i(\boldsymbol{q}-\boldsymbol{q}')\cdot\boldsymbol{P}} \langle\rho(\boldsymbol{P})\rangle = cN\delta_{\boldsymbol{q},\boldsymbol{q}'}$

c: concentration



Second order: $c(1-c)N + c^2N^2\delta_{\boldsymbol{q},\boldsymbol{q}'}$

Diagrammatic expansion

- For higher orders, **crossing** amplitudes

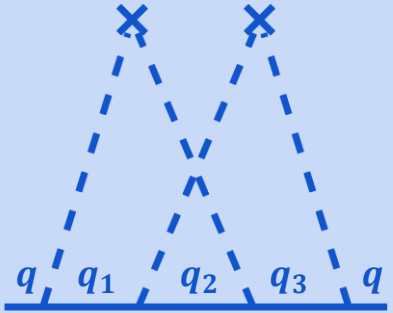
- An example, the 4th order crossing amplitude

$$q_1 - q + q_3 - q_2 = 0$$

Which corresponds to

$$\sum_{q_1 q_2} V_{q q_1} g_{q_1} V_{q_1 q_2} g_{q_2} V_{q_2, q+q_2-q_1} g_{q+q_2-q_1} V_{q+q_2-q_1, q}$$

- **Infinite** amplitudes → no **closed** solution

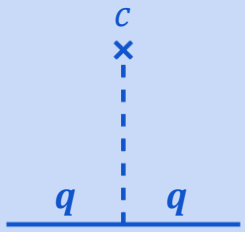


Virtual Crystal Approximation

- **VCA** decouples the average in the **potential**:

$$\langle VV \rangle \simeq \langle V \rangle \langle V \rangle$$
$$\langle T \rangle \simeq \langle V \rangle + \langle V \rangle g \langle V \rangle + \cdots$$

$$\Sigma_{qj} = cV_{qj}^{qj}$$



- V **hermitian** ⇒ Σ **real** ⇒ no scattering
- Σ **linear** in V ⇒ weak potentials

Average T-matrix Approximation

- ATA decouples the average in the **single-site** T-matrix:

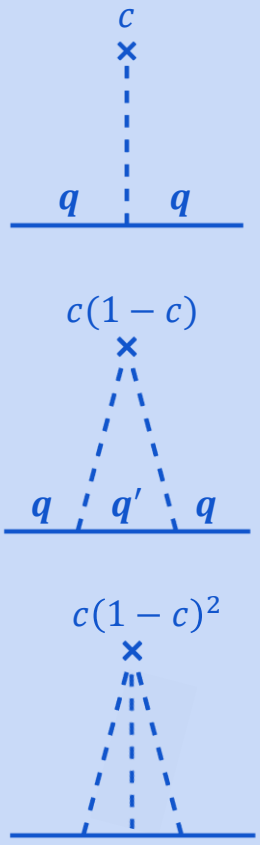
$$\langle T \rangle \simeq \langle T_1 \rangle + \langle T_1 \rangle g' \langle T_1 \rangle + \cdots$$
$$\langle T_1 \rangle = cV_1(1 - gV_1)^{-1}$$

$g' = g - g(\mathbf{0})$ in real space

- $\Sigma_{\mathrm{ATA}} = \langle T_1 \rangle - \langle T_1 \rangle \mathrm{g}(0) \langle T_1 \rangle + \cdots = \langle T_1 \rangle (1 + g(0)\langle T_1 \rangle)^{-1}$
$= cV_1(1 - (1-c)gV_1)^{-1}$

Apart from occupation correction, $\Sigma \simeq \langle T_1 \rangle$

- It's correct up to **linear** order in c, since crossing diagrams have $\mathcal{O}(c^2)$

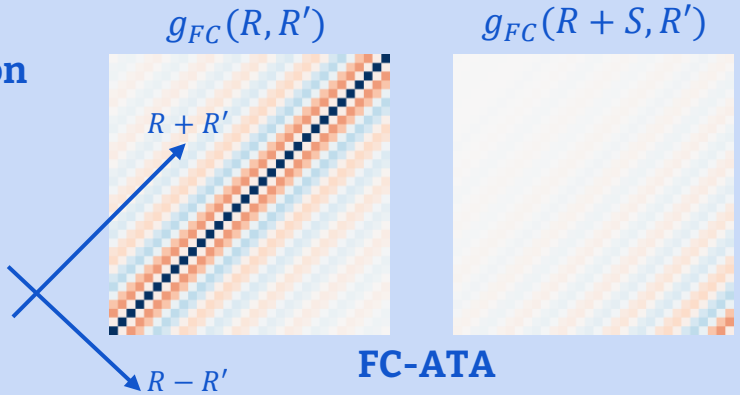


Average T-matrix Approximation

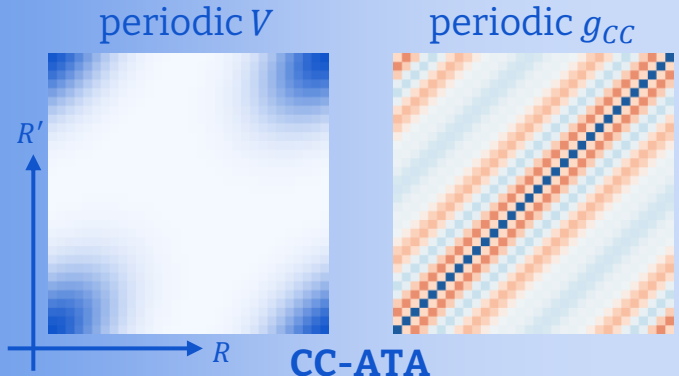
S: SC translation

- g_{CC} FT from **coarse** q-grid lacks spectral **resolution**
- $g_{FC}(S) \neq g_{FC}(0)$ makes it **ill-defined**
- g_{FF} is **projected** onto the $V \neq 0$ support

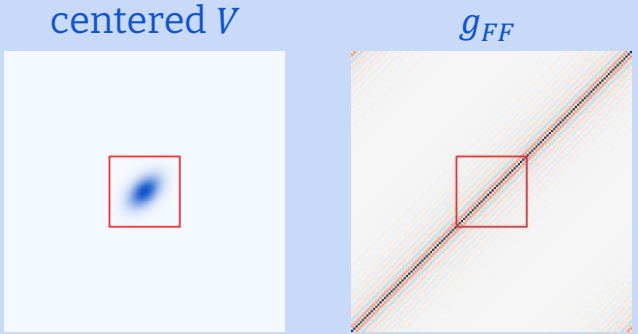
C: coarse
F: fine
FT: Fourier Transform



FC-ATA



CC-ATA



FF-ATA (our result)

Dynamical Cluster Approximation

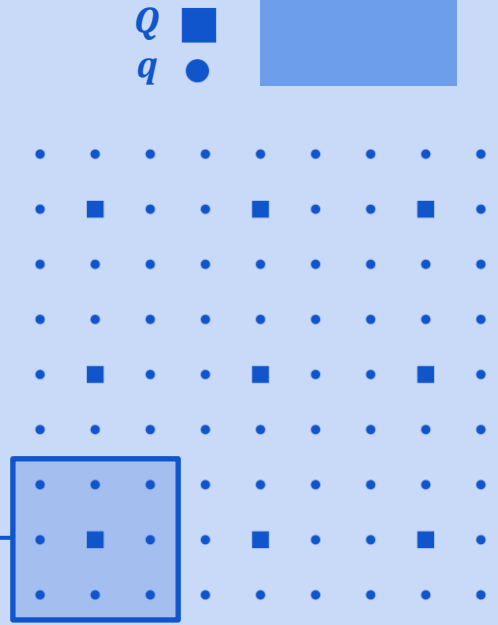
$$G = \langle (g^{-1} - V^{\alpha})^{-1} \rangle_{\alpha}$$
$$G^{-1} = g^{-1} - \Sigma$$

- DCA solves the defect problem **self-consistently** in a coarse Q-grid:

$$G_{n+1} = \left\langle \left(G_n^{-1} - (V^{\alpha} - \Sigma_n) \right)^{-1} \right\rangle_{\alpha}$$
$$\Sigma_{n+1} - \Sigma_n = G_n^{-1} - G_{n+1}^{-1}$$

- **Coarsening** operator is an integral in a patch centered at $\boldsymbol{Q}$:

$$G_n(\boldsymbol{Q}) = \int_{C(\boldsymbol{Q})} d\boldsymbol{q} \left(g_{\boldsymbol{q}}^{-1} - \Sigma_n(\boldsymbol{Q}) \right)^{-1}$$

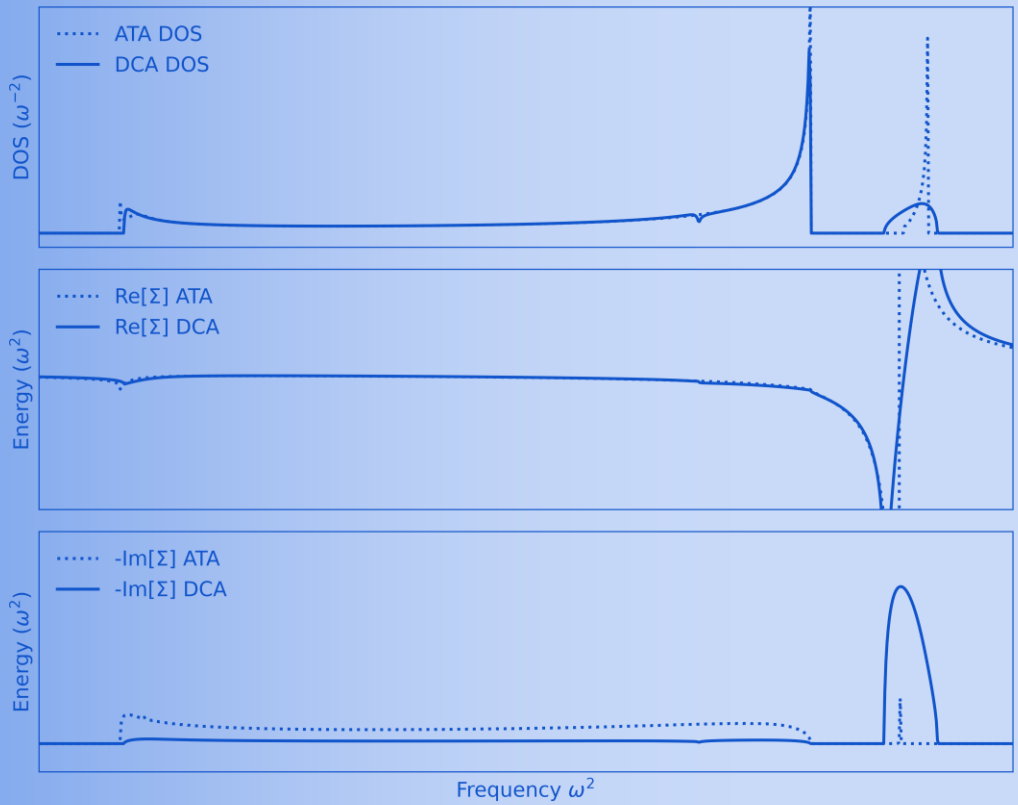


Dynamical Cluster Approximation

- Complex version of **Broyden's** mixing
- Average over a small **subset** of configurations
  Symmetries are enforced after the configuration average
- The **coarsening** operator can be inverted (Fourier interpolation)

$$\Sigma(\boldsymbol{Q}) \rightarrow \Sigma(\boldsymbol{q}), \qquad G(\boldsymbol{Q}) \rightarrow G(\boldsymbol{q})$$

Dynamical Cluster Approximation



1D system, with
$\Phi(R) \sim e^{-\beta R}$

Results

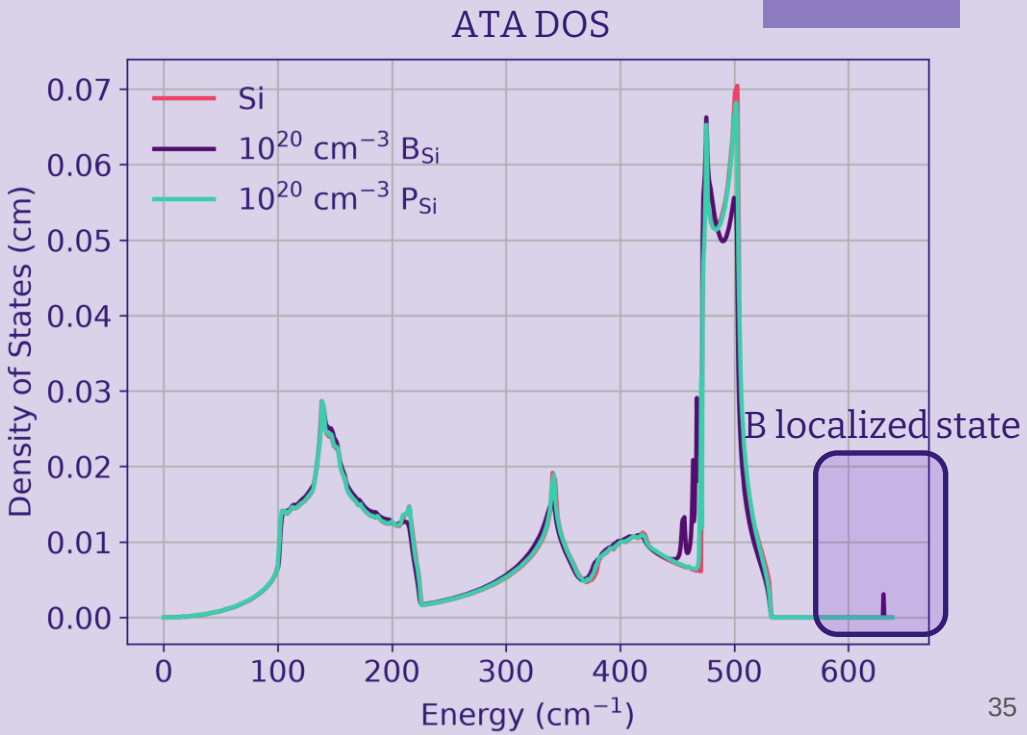
4.1 Computational details
4.2 P and B doping in Silicon
4.3 Te substitution in WSe_2
4.4 SiGe alloys

Computational details

- Electronic and IFC calculations are done in **Quantum ESPRESSO**
- My code, implemented in **D3Q** (anharmonic), takes Φ_0 and Φ as inputs
- Features:
  - automatic defect **centering**
  - V for **equivalent sites** through symmetry
  - Stable complex and symmetric version of **tetrahedron method** for g
  - **ATA/DCA** self-energy $\Sigma_{jj'}(\boldsymbol{q}, \omega)$
  - Linear **interpolation** of linewidth $-2\Im\Sigma(\boldsymbol{q}, \omega_{\boldsymbol{q}j})$

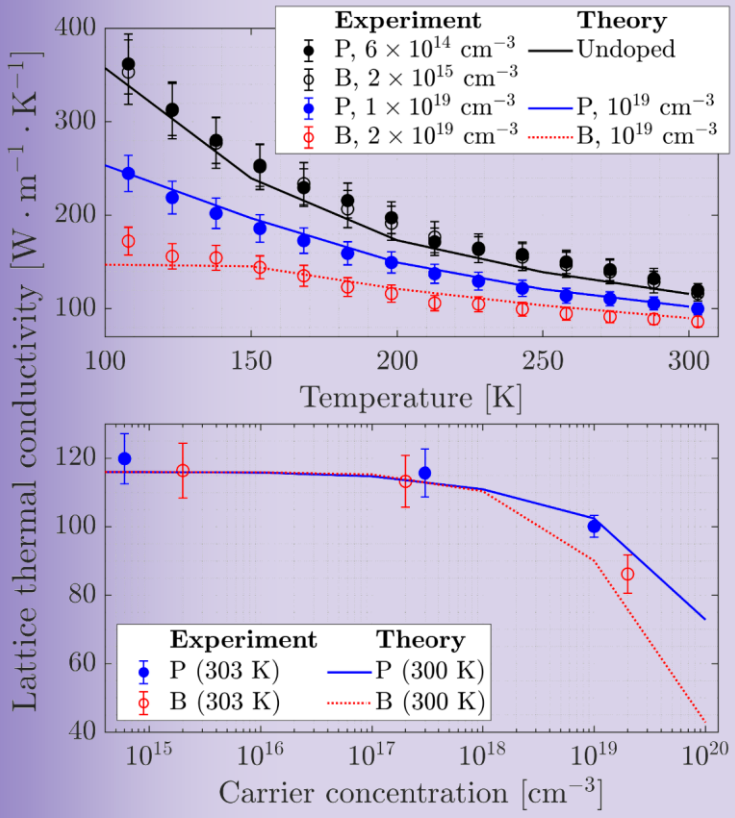
P and B doping Silicon

- Electrical and thermal transport depends on doping
- How doping modifies thermal conductivity?

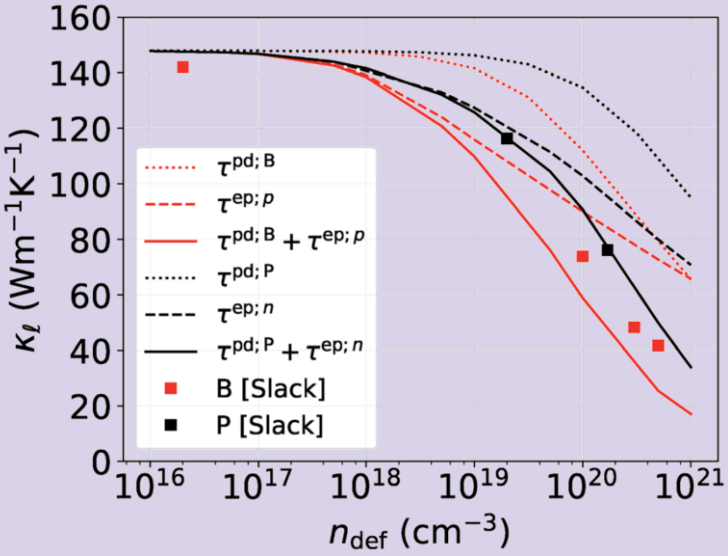


P and B doping Silicon

Sen et Al (2026) submitted PRB

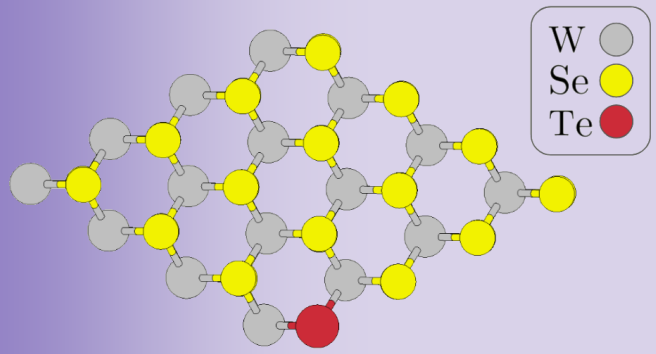


- With 3-phonons and electron-phonon scattering, we can calculate the **lattice** thermal conductivity
- Phonon-defect scattering is **FC-ATA**



Dongre et al. (2020), J. Mater. Chem. A 8, 1273

Te substitution in WSe_2

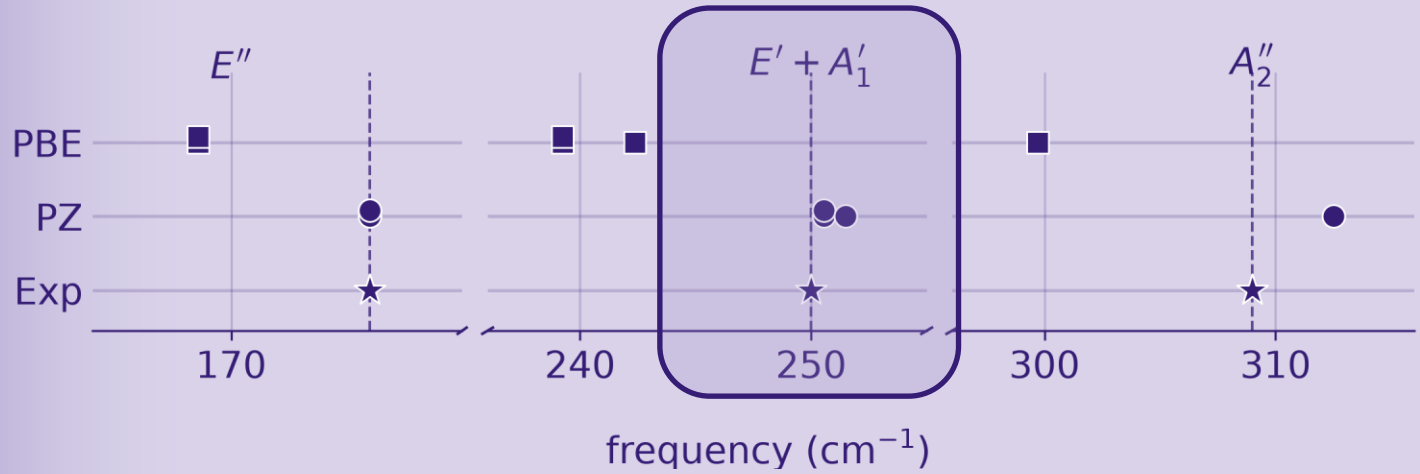


- **2D** material, monolayer of Se-W-Se planes

  15 Å slab of vacuum, electrostatic cutoff

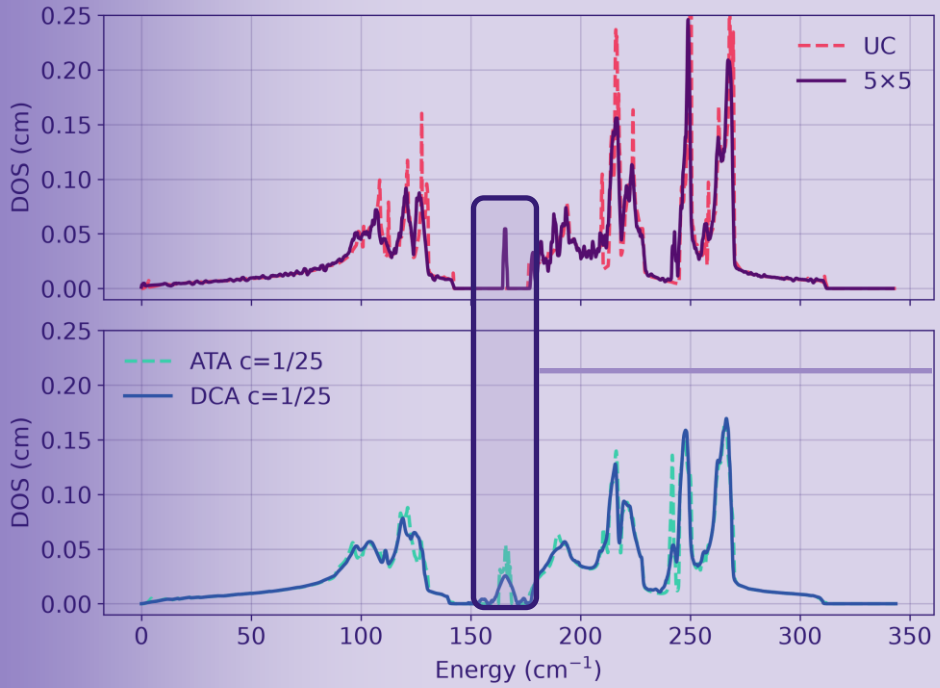
- Te substitues to Se in a 5 × 5 grid

- PZ functional approximates better phonons at Γ

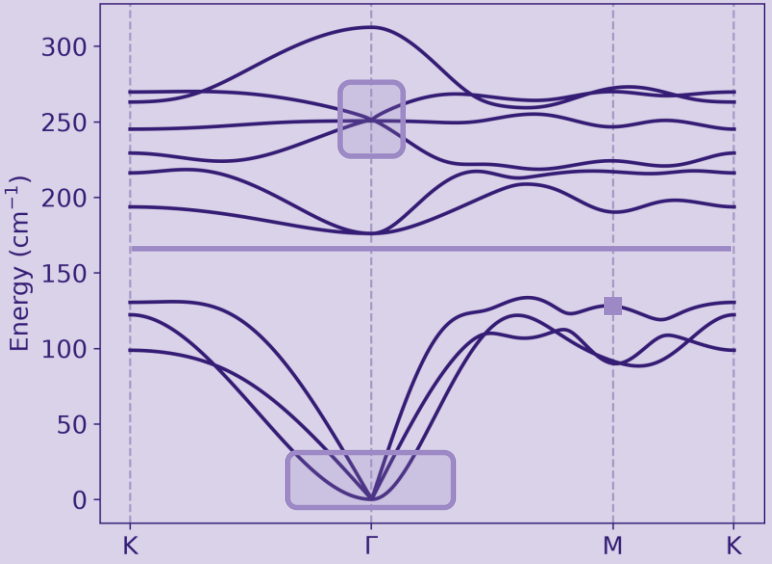


Te substitution in WSe_2

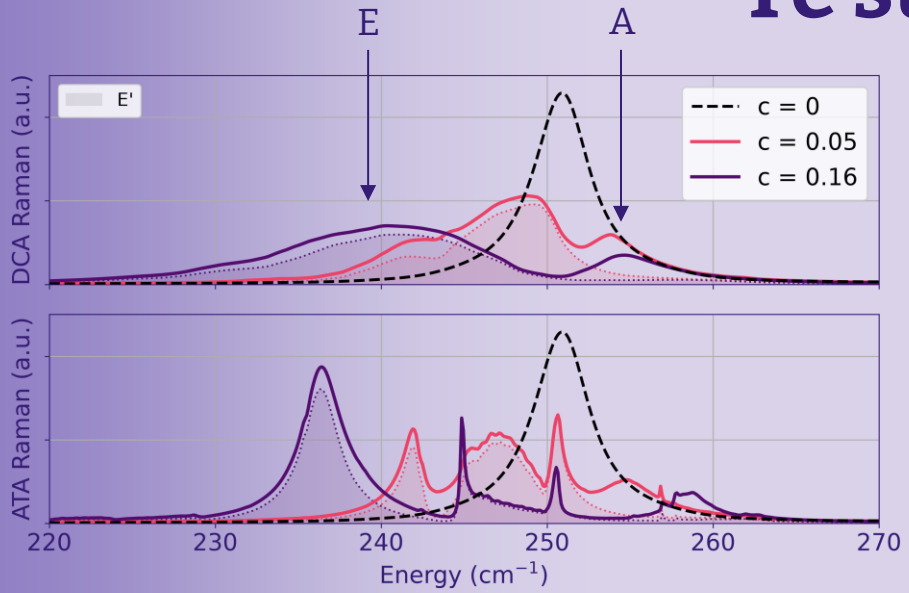
- One **quadratic** acoustic phonon branch at Γ



- ATA and DCA broaden the **localized state** at 170 cm^{-1}



Te substitution in WSe_2

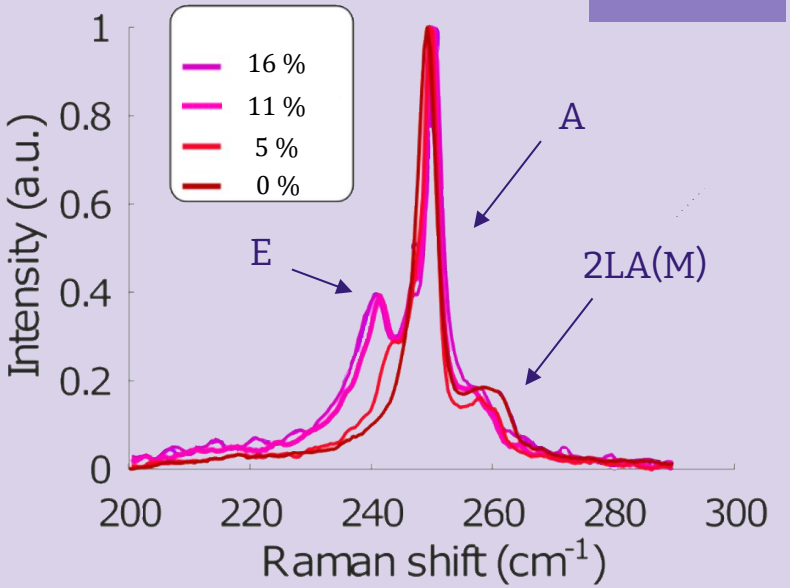


- Raman from Γ **spectral function** (3ph + ph-def) and **Raman intensities** from 2° order displacements

- 2LA(M) is **higher** order Raman:

$$|p_{in}\rangle \rightarrow |p_{out}\rangle + |-\boldsymbol{M}\rangle + |\boldsymbol{M}\rangle$$

Di Bernardino submitted PRB



SiGe alloys

- For alloys, we start from **VCA** to better interpolate the concentration between 0 and 1

$$\Phi_0 = (1-c)\Phi_{Si} + c\Phi_{Ge}$$

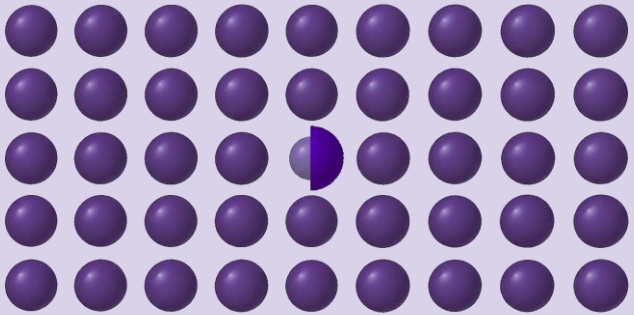
The perturbations $\Delta\Phi$ must start from Φ_0

- Lattice constant, frequencies, velocities and anharmonic linewidths are linearly **interpolated**

- The T-matrix averages the two contributions

$$\langle T \rangle = (1-c)t[V_{Si}] + ct[V_{Ge}]$$

DCA uses $V_{Si} - \Sigma$ and $V_{Ge} - \Sigma$ in the cluster solver



SiGe alloys

$$V\bar{M} = \Delta\Phi - \omega^2 \Delta M$$

- The mass term already **satisfies** the VCA condition:

$$(1-c)\frac{\Delta M_{Si}}{\bar{M}} + c\frac{\Delta M_{Ge}}{\bar{M}} = 0$$

- Perturbations are **fluctuations** from the average:

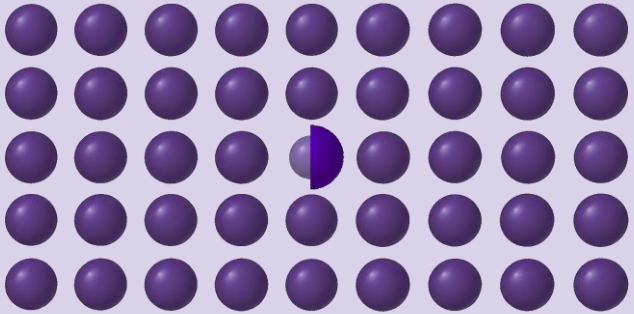
$$(1-\mathrm{c})\Delta\Phi_{Si} + c\Delta\Phi_{Ge} = 0$$

- They must decay away from defect:

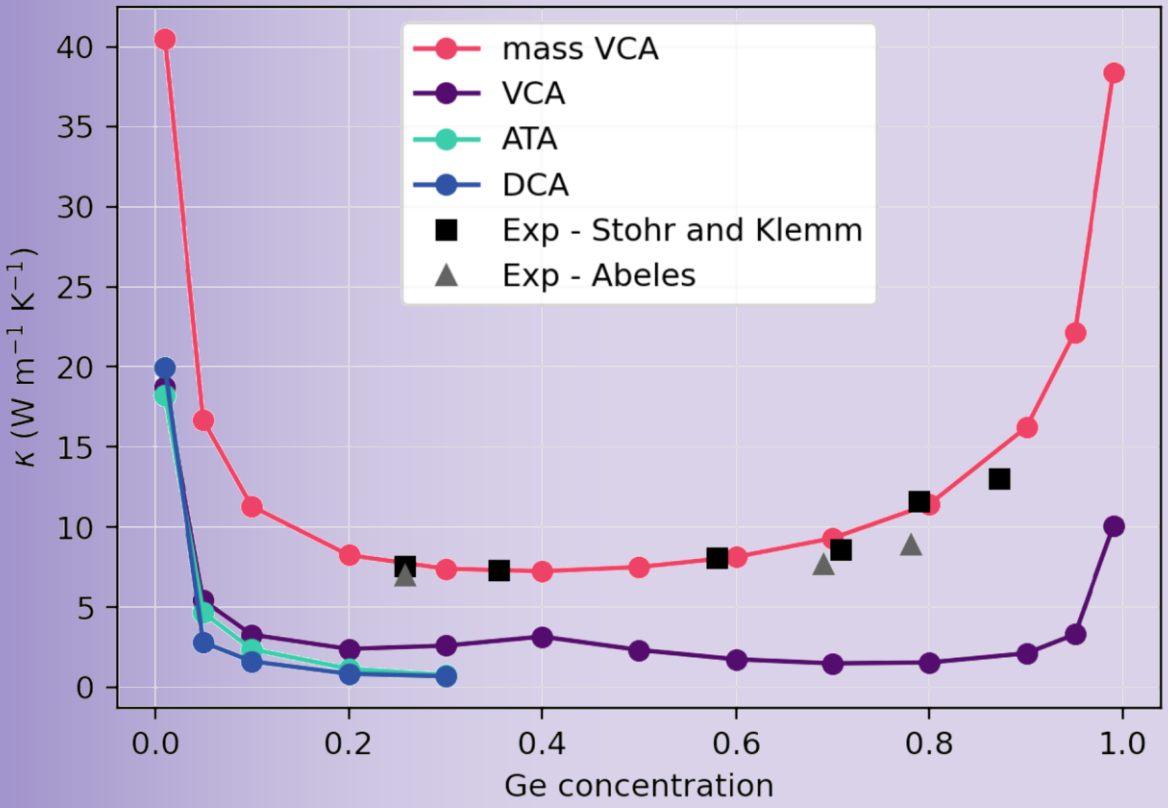
$$\Delta\Phi \to 0$$

- They must recover the $c \to 0$ limits:

$$\Delta\Phi_{Ge} \to \Phi_{Ge|Si} - \Phi_{Si}$$

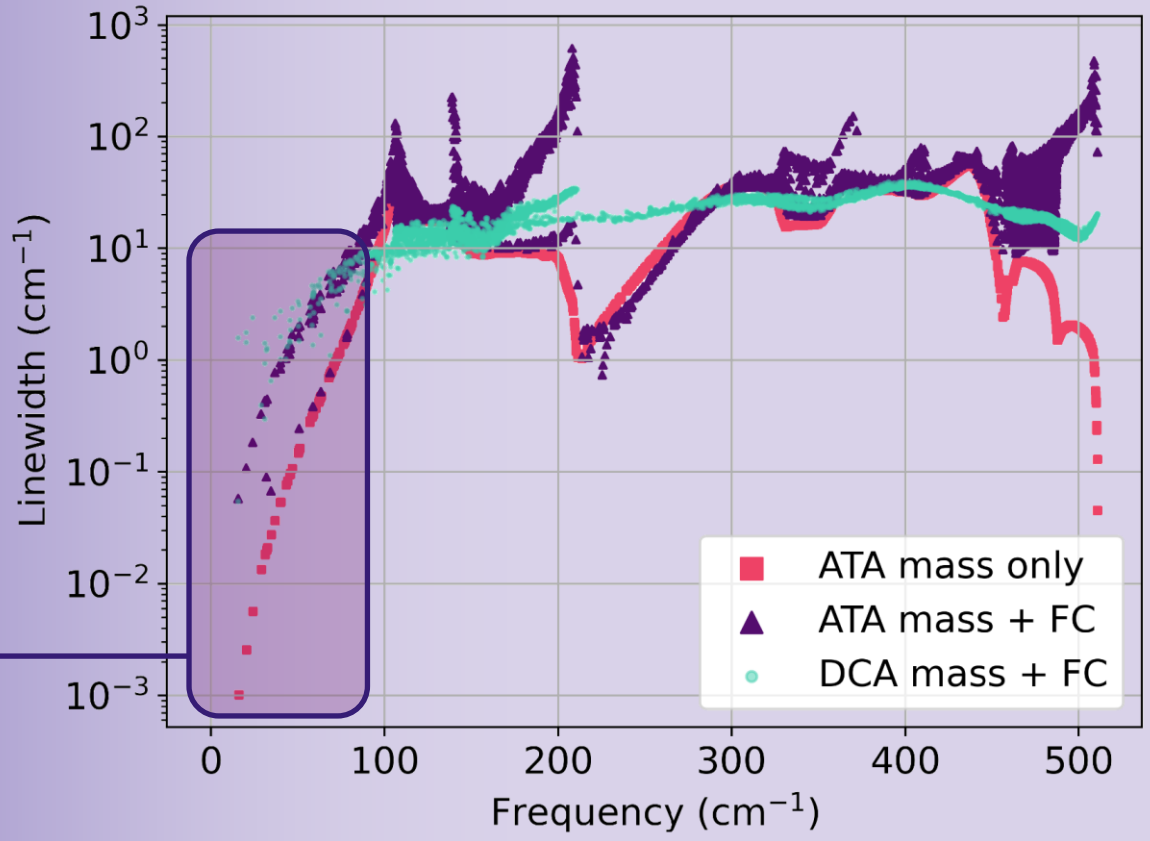


SiGe alloys


mass VCA
VCA
ATA
DCA
Exp - Stohr and Klemm
Exp - Abeles
κ (W m⁻¹ K⁻¹)
Ge concentration
0
5
10
15
20
25
30
35
40
0.0
0.2
0.4
0.6
0.8
1.0

SiGe alloys

$Si_{0.9}Ge_{0.1}$



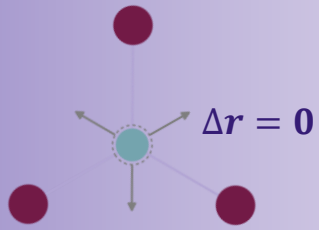
Linewidth $= C\omega^4$

SiGe alloys

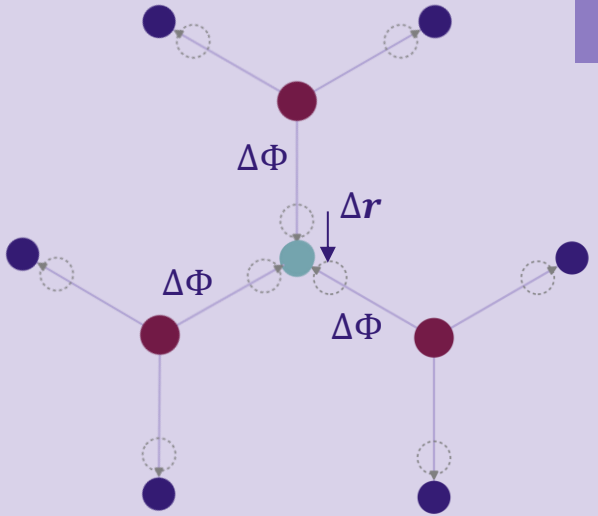
- $\sum_P \Delta\Phi_1(\boldsymbol{P})$ overestimates the real perturbation

  Relaxation $\rightarrow \Delta\boldsymbol{r} \rightarrow \Delta\Phi \sim e^{-\beta\Delta r}$

$\Delta\Phi_{\mathrm{TOT}} \simeq 0$



$\Delta\Phi_{\mathrm{TOT}} = 3\Delta\Phi$



Ge
Si
Si

Future perspectives

- Use $\sum \Delta \boldsymbol{r}$ information to fit a better $\Delta\Phi^{\alpha}$
- Use a material-specific defect distribution in DCA
- Green-Kubo $\langle \kappa \rangle \sim \langle GG \rangle \neq \langle G \rangle^2$

Acknowledgements

- École Doctorale Physique Ile de France (EDPIF) fellowship
- ANR-21-CE50-0008 PlaCHO
- IDRIS supercomputing GENCI grant 7320

Thank you!

Contents

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2. **Defective potential**
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3. **Disorder approximations**
   - 3.1 Diagramatic expansion
   - 3.2 Virtual Crystal Approximation (VCA)
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4. **Results**
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Tetrahedron weights

GF are often evaluated near their **singularities**, where

$$(\omega + i0^+ - \omega_0)^{-1} = \mathcal{P}\left((\omega - \omega_0)^{-1}\right) - i\pi\delta(\omega - \omega_0)$$

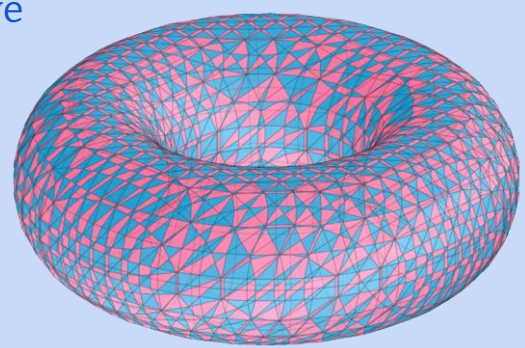
$\mathcal{P}$ and δ are principal value and Dirac delta **distributions**, they are **integrable**

GF are calculated as integrals in a small region around the q-point:

$$g_{qj}(\omega) = \int dq' \left(\omega^2 - \omega_{q'j}^2\right)^{-1}$$

Integrals of **linearly interpolated** ω_{qj} over **tetrahedra** have an analytical formula, even for **complex** energies

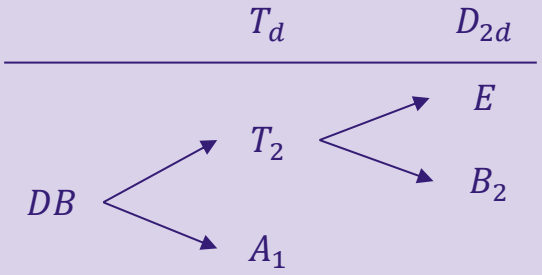
Symmetry can be exploited to reduce calculations in the **irreducible** Brillouin Zone



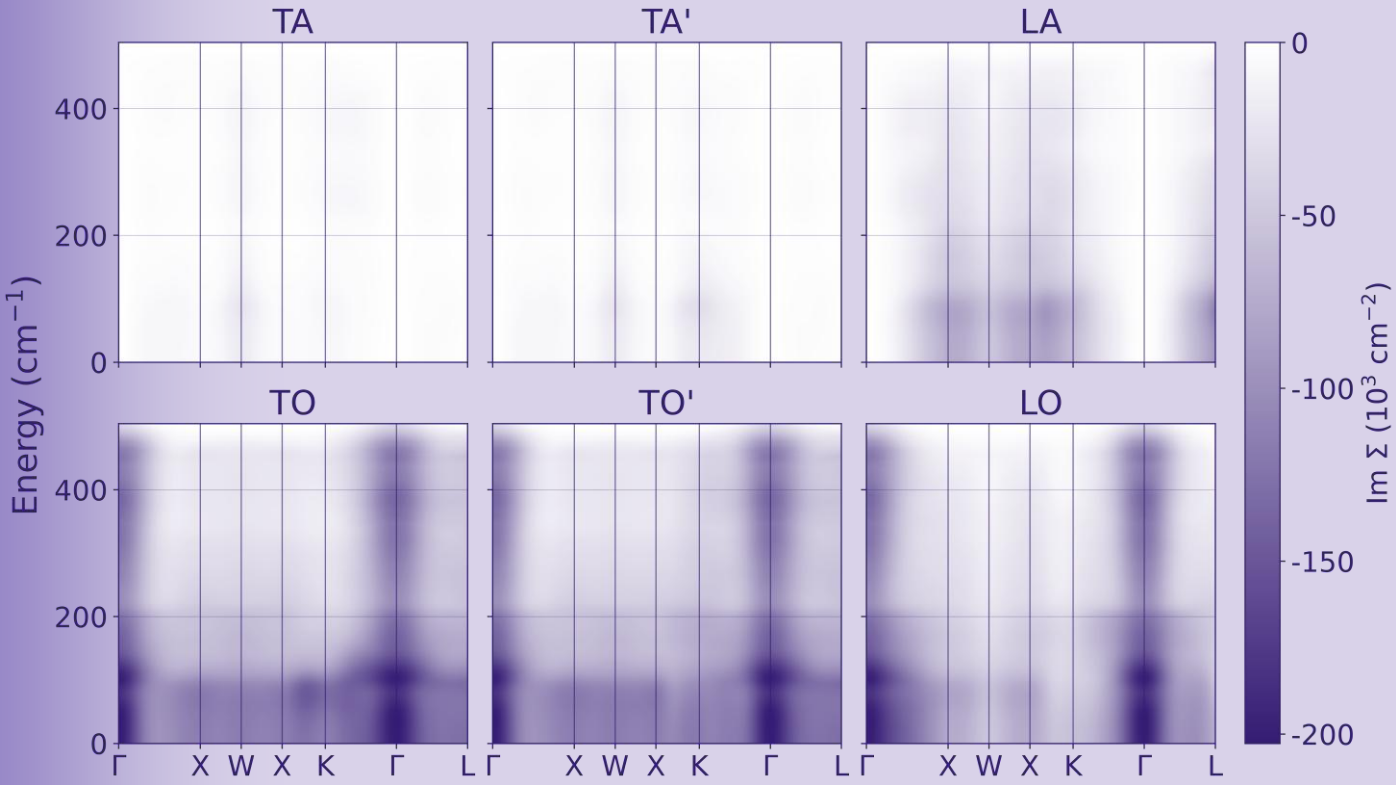
Vacancies in Silicon

Vacancies are treated by setting $\Phi_{dj} = M_d = 0$ in the coordinates d where the vacancy lie

Silicon vacancy creates 4 dangling bonds in a tetrahedral environment T_d, Jahn-Teller distortions reduce the symmetry to D_{2d}

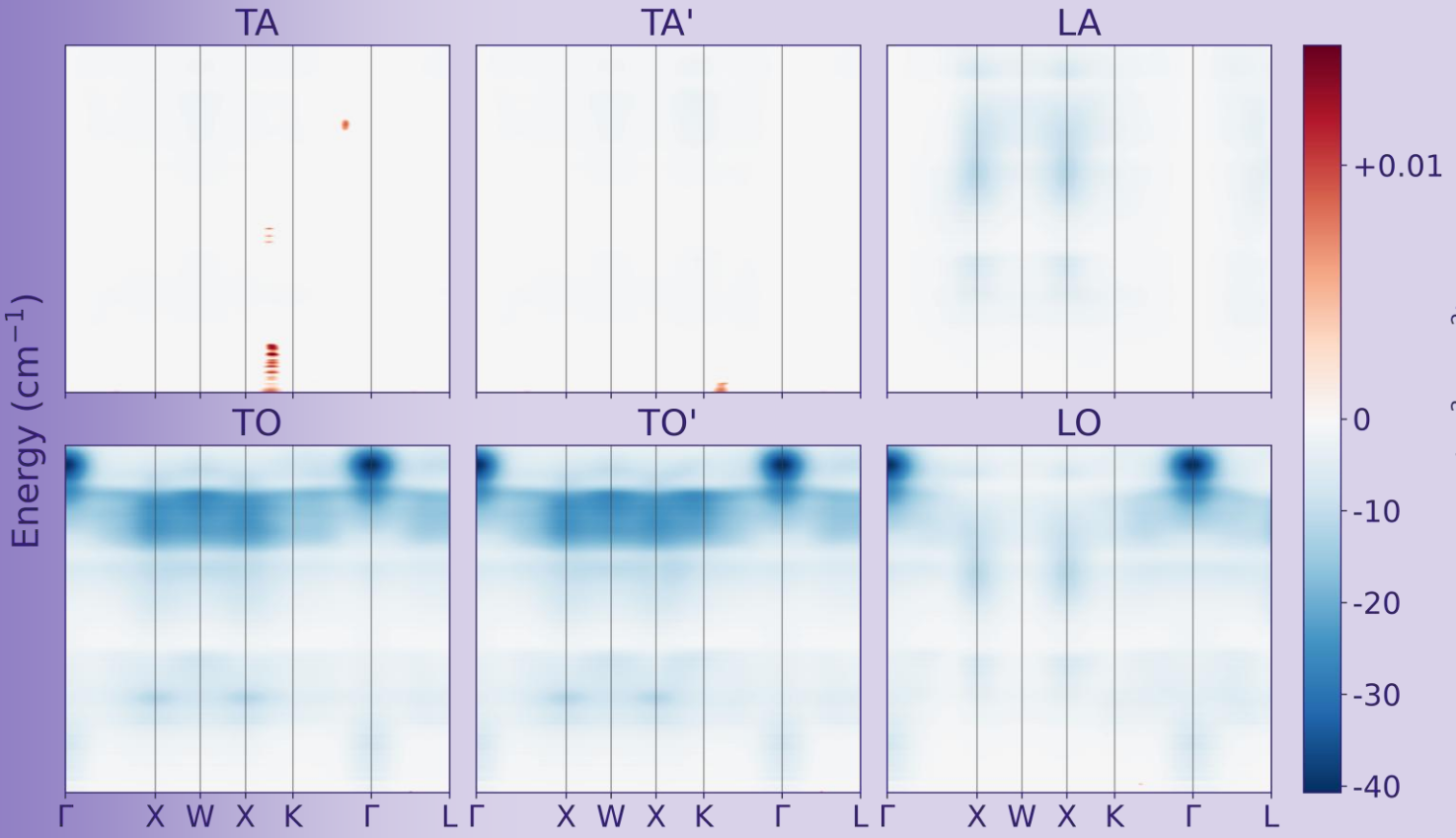


Vacancies in Silicon



DCA grid $4 \times 4 \times 4$
$c = 5\%$

P and B doping Silicon



DCA grid $4 \times 4 \times 4$

$c = 5\%$

Computational details

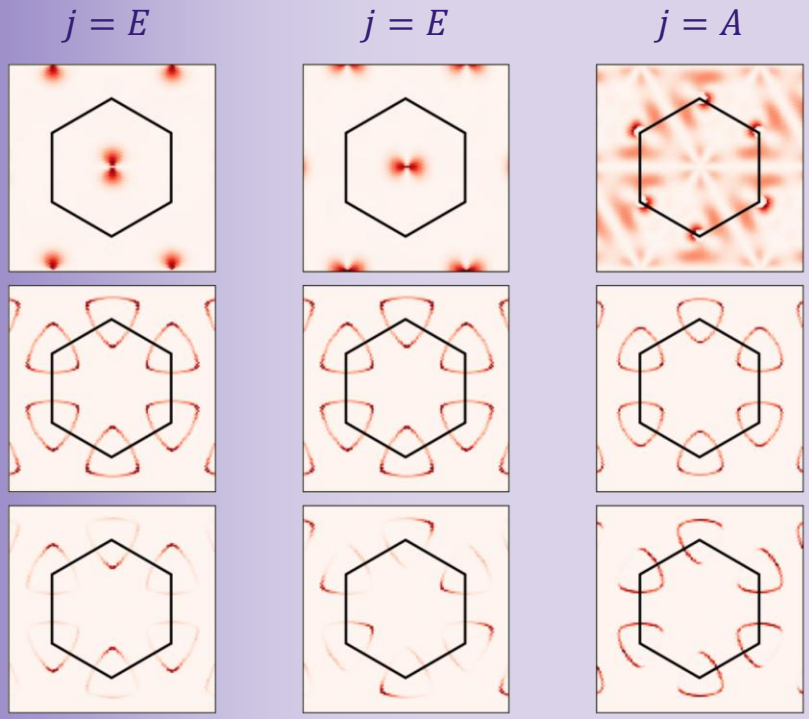
QE parameters

pseudo	NC
functional	PZ/PBE
k-points	$8 \times 8 \times 8$
Gauss smearing	10^{-3}
SCF threshold	10^{-12}
energy threshold	10^{-4}
force threshold	10^{-3}
q-points	$4 \times 4 \times 4$
PH scf[2]	10^{-16}

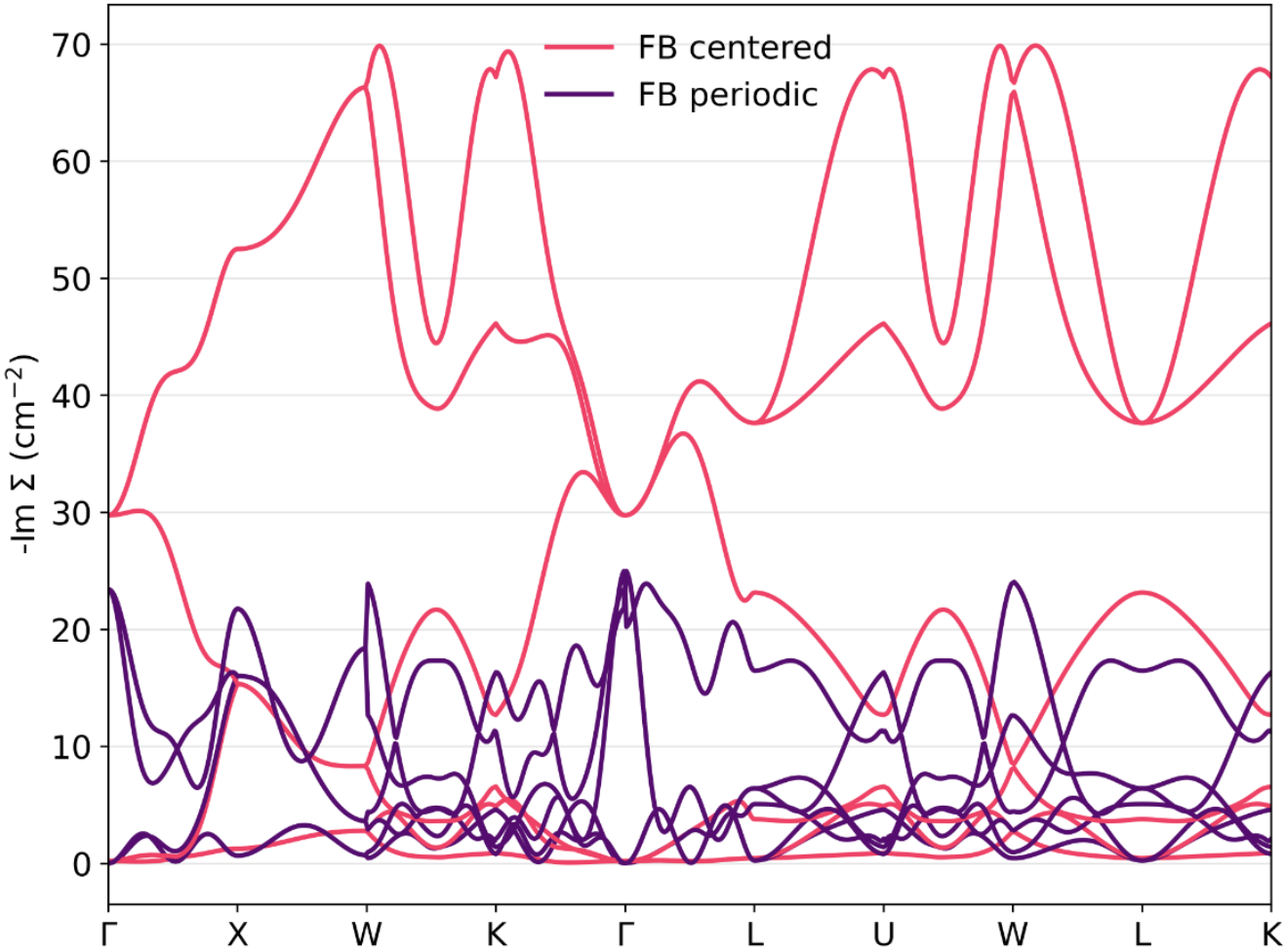
my parameters

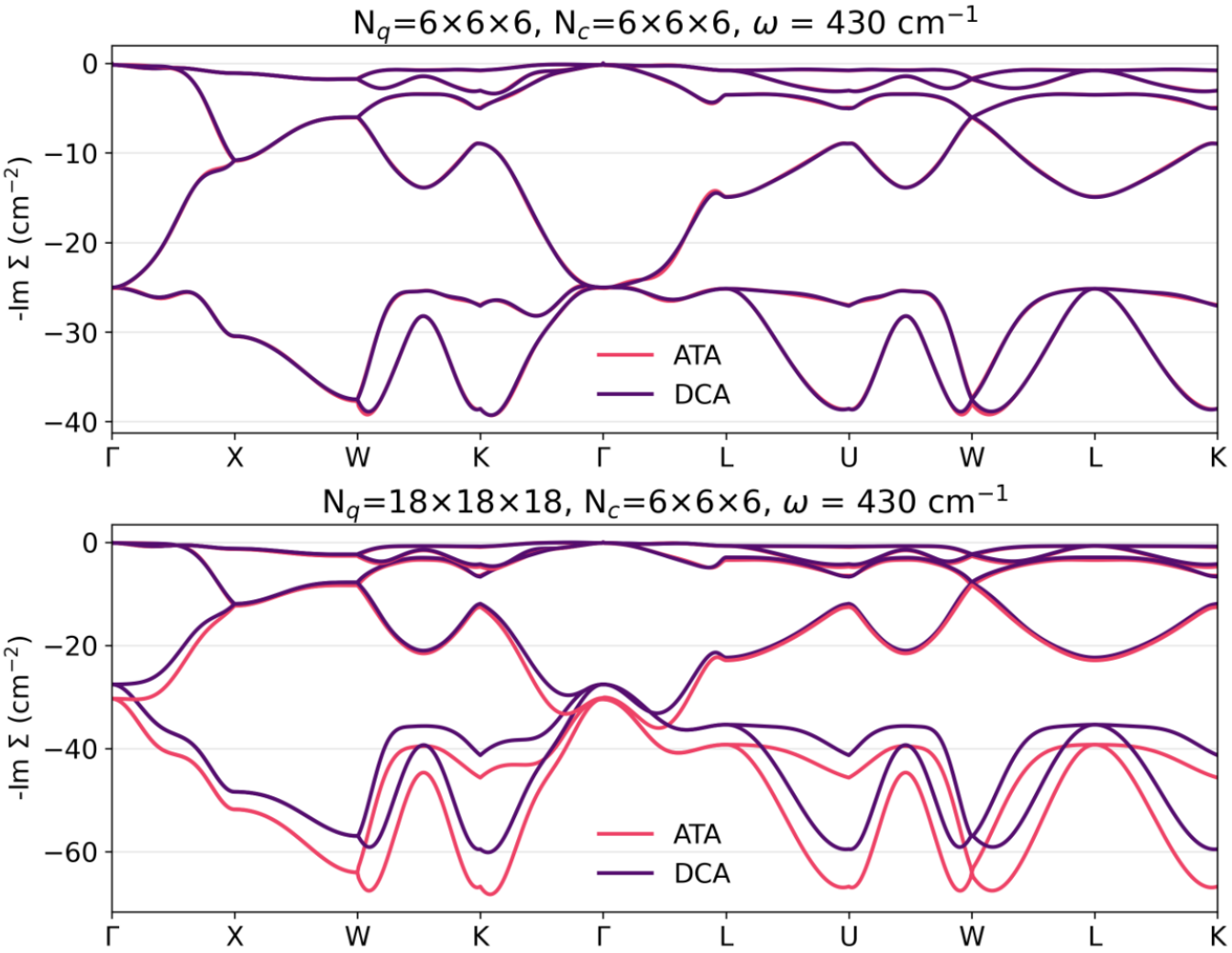
concentration	$< 10\%$
inner grid	$< 40 \times 40 \times 40$
ω interval	$100 - 1000$
DCA grid	$< 6 \times 6 \times 6$
DCA samples	$50 - 200$

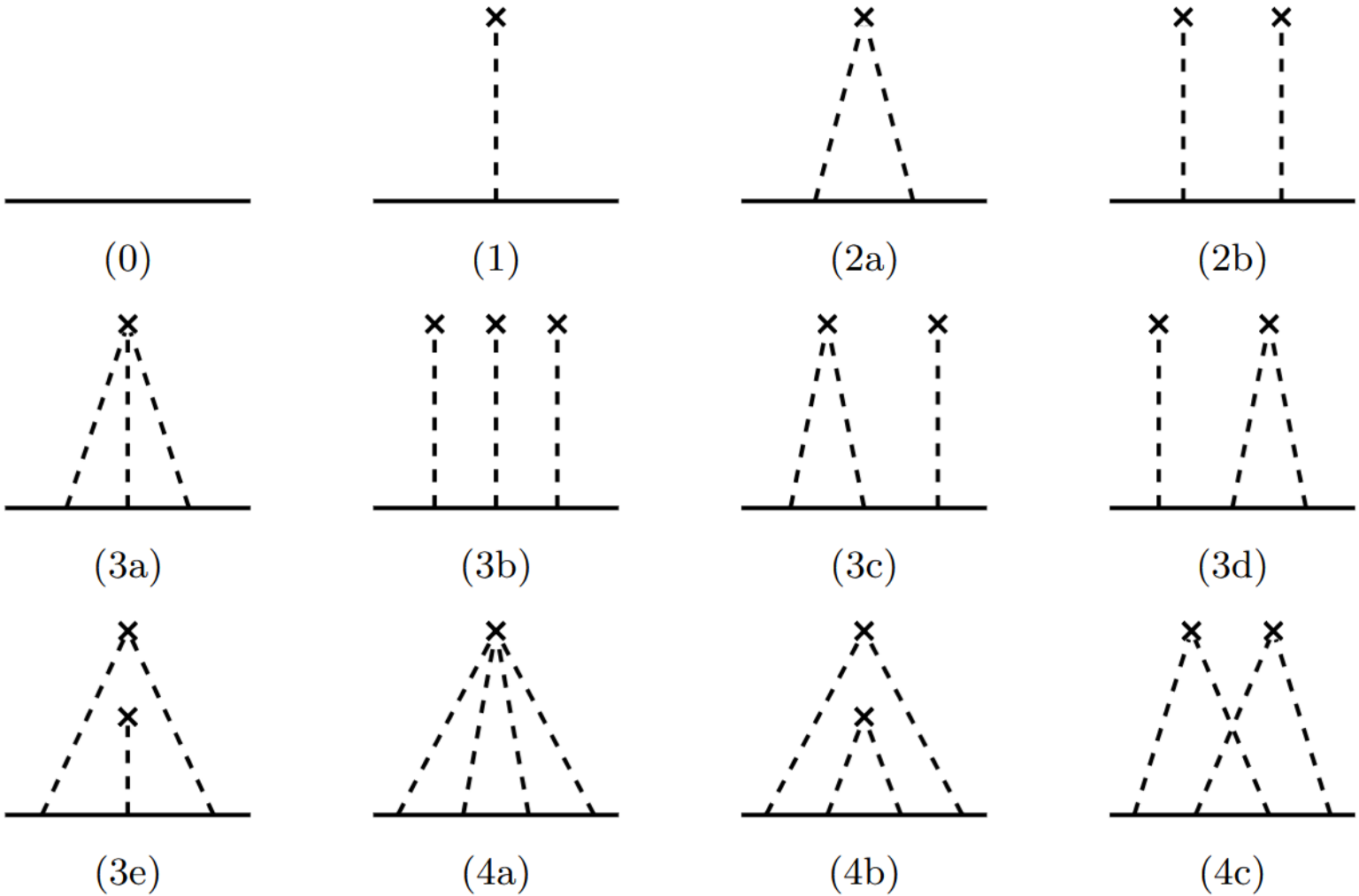
Te substitution in WSe_2



- How the perturbation looks like?


FB centered
FB periodic
-Im Σ (cm^{-2})
70
60
50
40
30
20
10
0
Γ
X
W
K
Γ
L
U
W
L
K


N_q=6×6×6, N_c=6×6×6, ω = 430 cm^{-1}
-Im Σ (cm^{-2})
0
−10
−20
−30
−40
ATA
DCA
Γ
X
W
K
Γ
L
U
W
L
K
N_q=18×18×18, N_c=6×6×6, ω = 430 cm^{-1}
-Im Σ (cm^{-2})
0
−20
−40
−60
ATA
DCA
Γ
X
W
K
Γ
L
U
W
L
K


(0)
(1)
(2a)
(2b)
(3a)
(3b)
(3c)
(3d)
(3e)
(4a)
(4b)
(4c)

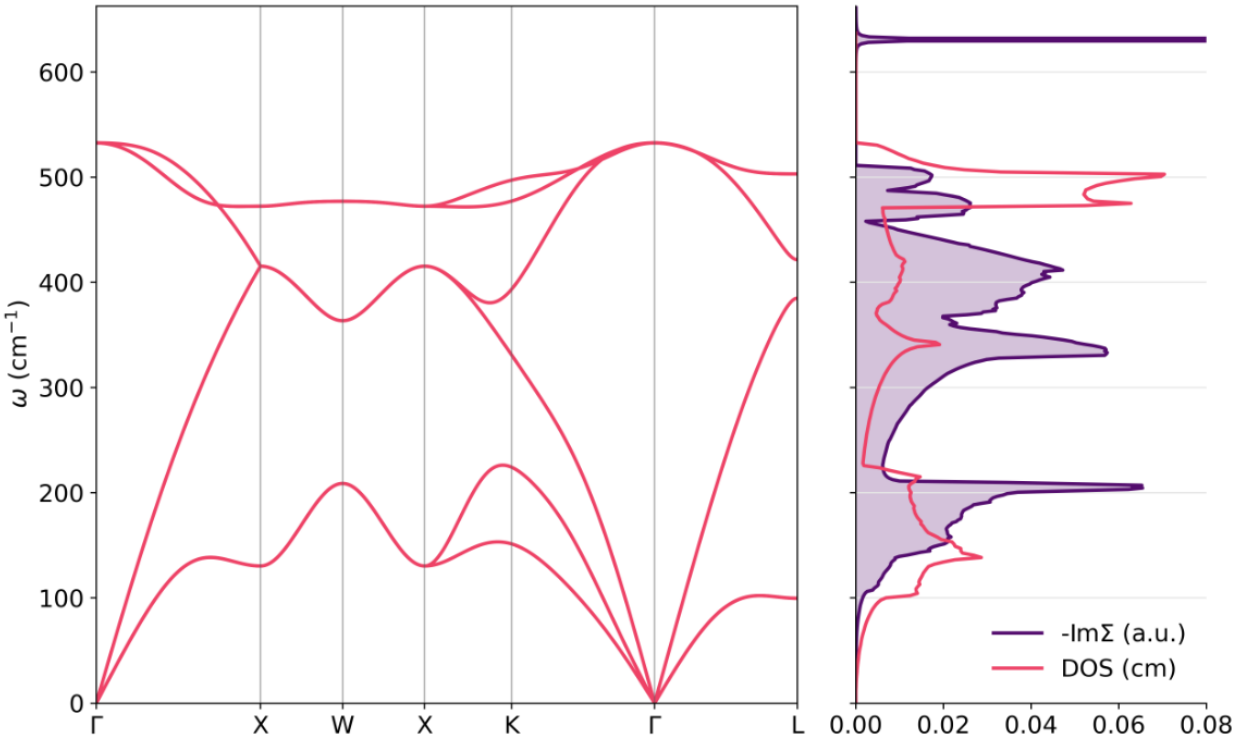
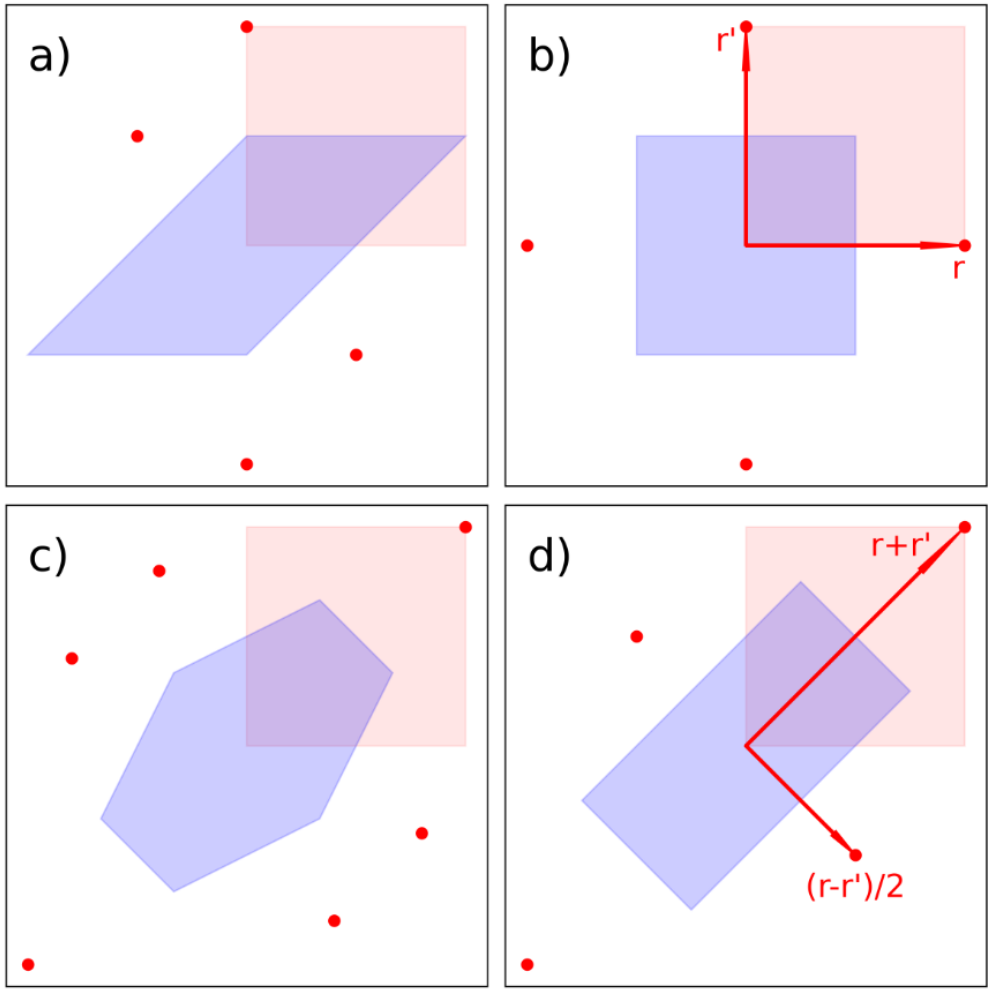


Figure 1.6: *Energy width for single mass defect in Silicon, with* $\Delta m/m = -0.6$. *The mass defect is maximally localized, hence the width is expected to be constant in reciprocal space, for cubic crystals. The lines up to* ~ 500 *are inside the DOS of the system, while the line at* ~ 630 *is the localized state, outside the bands. For phonons,* $E = \omega^2$ *and the equation to solve is* $\omega^2 = \omega_0^2 + 0.6\omega^2$, *where* ω_0 *is the center of the band.*


a)
b)
r'
r
c)
d)
r+r'
(r-r')/2

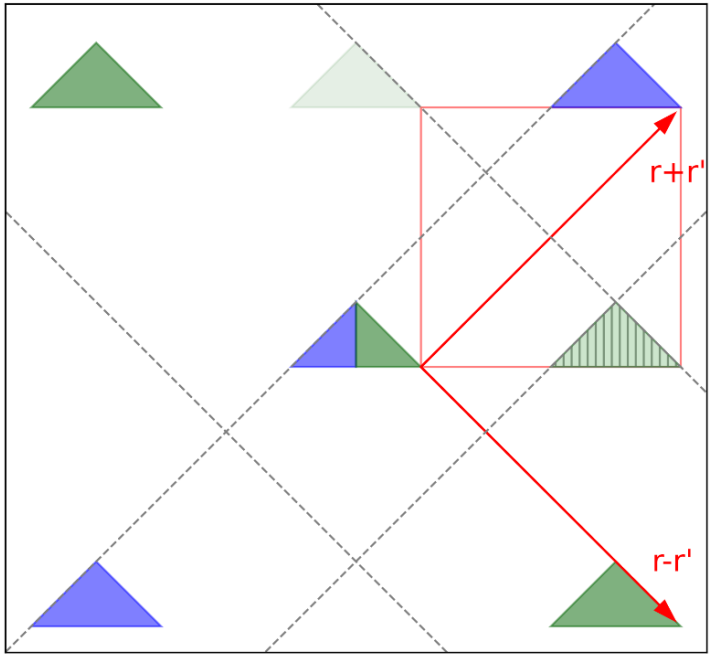
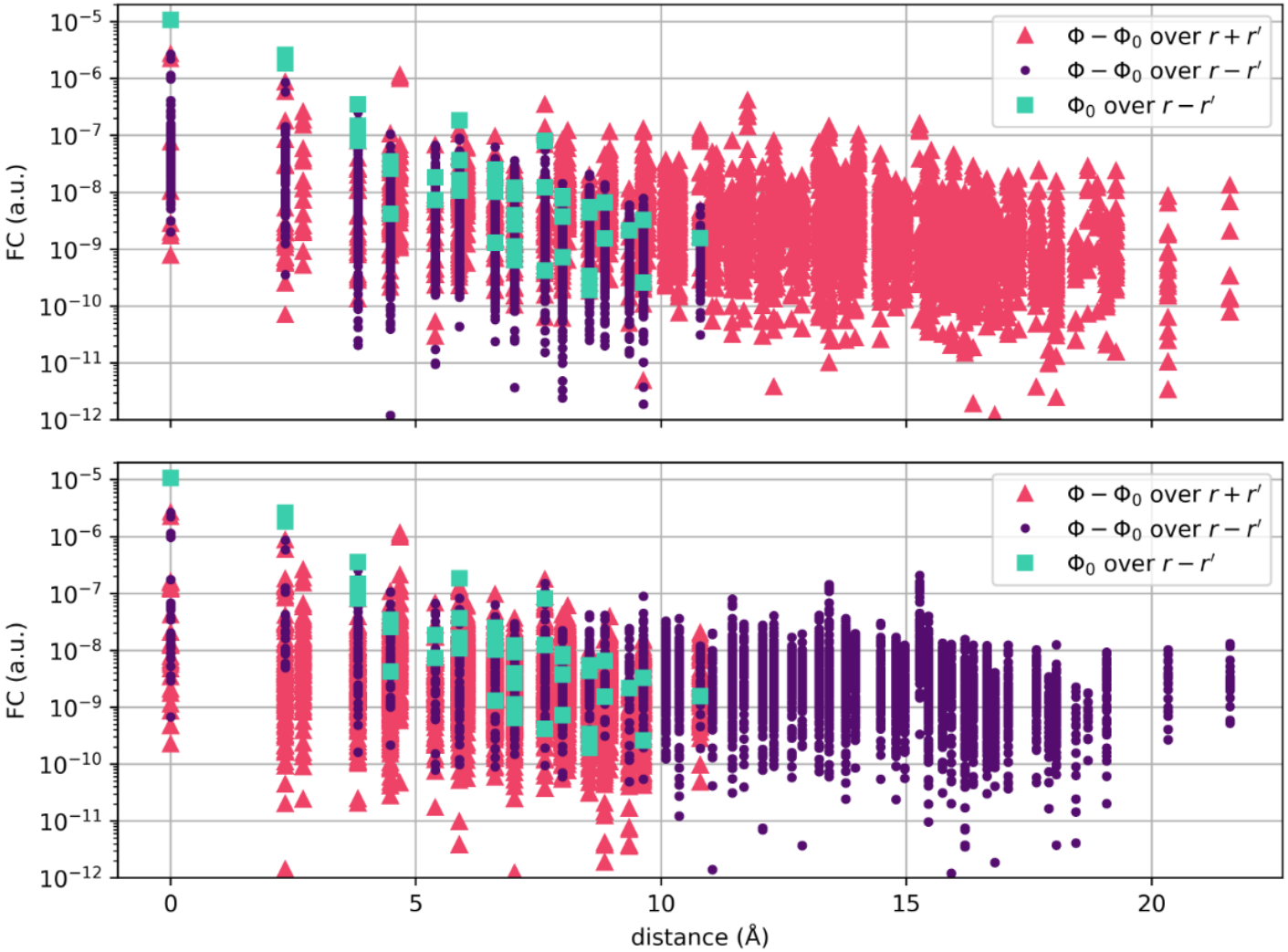


Figure 1.3: *Showcase of centering method d) for a one-dimensional system. The red square delimits the supercell's area, the hatched triangle is to be shifted according to the algorithm. This triangle is outside the centered Wigner-Seitz cell for the difference, but not for the sum. For the sum, the two classes of solutions are depicted in light green for* $\boldsymbol{T}_+ = 0$ *and dark green for* $\boldsymbol{T}_+ = -1$*. The only class of solutions for the difference are blue and correspond to* $\boldsymbol{T}_- = -1$*. They intersect for* $\boldsymbol{T} = -1$ *and* $\boldsymbol{T}' = 0$*, which is the only solution for which both* $\boldsymbol{T}$ *and* $\boldsymbol{T}'$ *are lattice vectors. In the figure, the centered triangle is dark green and blue.*


$\Phi - \Phi_0$ over $r + r'$
$\Phi - \Phi_0$ over $r - r'$
Φ_0 over $r - r'$
FC (a.u.)
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
0
5
10
15
20
distance (Å)